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A Decomposition of 10-K Risk-Factor Language and Post-Filing Stock Return Volatility

Bachelor Thesis

Study programme: Data Analytics

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Abstract

The risk-factor section (Item 1A) of the 10-K describes the risks a firm faces. This thesis asks whether three properties of the section are associated with stock-return volatility in the weeks after the filing: its length (H1), its share of words from the Loughran–McDonald uncertainty and litigious lists (H2), and the share of its industry peers' vocabulary that the firm does not use (H3). The panel of 5,782 firm-years of 512 large U.S. non-financial firms for fiscal years 2010–2024 is built directly from SEC filings. Regressions of log 30-day post-filing volatility control for pre-filing volatility, size, valuation, past returns and accounting fundamentals, with industry and fiscal-year fixed effects and standard errors clustered by firm and year. Longer Item 1A sections are associated with higher post-filing volatility ($\hat{\beta} = +0.053$, $t = +4.51$) at every horizon from 5 to 365 days, but the association does not hold within firms over time. Holding length fixed, the density of risk words is not significant at 30 days; only the litigious words have a negative association, which is sensitive to the sample. The peer-relative omission gap is not associated with volatility. H1 is supported between firms; H2 and H3 are not supported.

Keywords

textual analysis, 10-K filings, SEC EDGAR, volatility, risk factors, Item 1A, Loughran–McDonald, LM-Litigious word list, boilerplate disclosure, panel data, corporate disclosures

Contents

Introduction 6

1 Literature Review 7

2 Data 8

2.1 Sample selection . . . 10
2.2 Annual filings as the unit of observation . . . 10
2.3 Financial data from XBRL . . . 12
2.4 Text extraction from 10-K filings . . . 14
2.4.1 Source of the text . . . 14
2.4.2 Section identification and exhibit fallback . . . 14
2.4.3 Post-processing and tokenisation . . . 15
2.4.4 Coverage . . . 16

3 Methodology 17

3.1 Outcome variable and timing . . . 17
3.2 Control variables . . . 19
3.3 H1 — Item 1A length and disclosure volume . . . 19
3.4 H2 — Risk-word density . . . 20
3.5 H3 — Peer-relative omission gap . . . 21
3.6 Estimation and inference . . . 23

4 Results 25

4.1 Descriptive statistics . . . 25
4.2 Length and risk-word density (H1 and H2) . . . 26
4.3 The omission gap (H3) . . . 31

5 Robustness 32

5.1 Alternative horizons . . . 32
5.2 Robustness of the H1 and H2 estimates . . . 33
5.3 Firm fixed effects . . . 36
5.4 The omission measure (H3) . . . 38
5.5 Persistence and unit roots . . . 41

Conclusion 43

References 45

A Data Construction and Supplementary Tables 48

A.1 Sample construction and coverage . . . 48
A.2 XBRL tags . . . 50
A.3 Filing-frequency pilot tests . . . 50

A.4 Additional robustness tables 52

B Documentation of AI Tool Usage **54**

List of Abbreviations

API Application Programming Interface
CIK Central Index Key
DV Dependent Variable
EDGAR Electronic Data Gathering, Analysis, and Retrieval
EPS Earnings Per Share
FE Fixed Effects
FY Fiscal Year
FYE Fiscal Year End
GLM Generalized Linear Model
L&M Loughran and McDonald
LLM Large Language Model
MD&A Management's Discussion and Analysis
NCI Non-Controlling Interest
NLP Natural Language Processing
OCF Operating Cash Flow
OLS Ordinary Least Squares
ROA Return on Assets
SEC Securities and Exchange Commission
SIC Standard Industrial Classification
TOC Table of Contents
XBRL eXtensible Business Reporting Language

Introduction

1. Literature Review

2. Data

This chapter describes how the firm-year panel is built. Financial variables come from the SEC EDGAR Company Facts API, the narrative sections are parsed from the HTML of the filings on EDGAR, and return-based measures are computed from daily prices downloaded from Yahoo Finance with the `yfinance` package. No commercial research database (such as Compustat, CRSP or a vendor 10-K corpus) is used. Building the data from these sources means that the textual and financial data can be matched to the same filing, period and reporting entity, and that the processing steps are visible in the code that accompanies the thesis. It also means that problems that commercial providers usually handle (XBRL tag heterogeneity, irregular HTML, amendments, share counts of multi-class firms) are dealt with directly; this chapter describes those choices and their consequences. Yahoo Finance prices are not raw: the adjusted close used for returns is adjusted for splits and dividends by the provider, and the price series were downloaded once and kept fixed for all computations. The definition, source and timing of every variable are listed in Appendix A.

Figure 2.1 gives an overview. The pipeline starts from the SEC sources, applies the sample filters, and runs a financial (XBRL) branch and a textual branch that are merged into the firm-year panel.

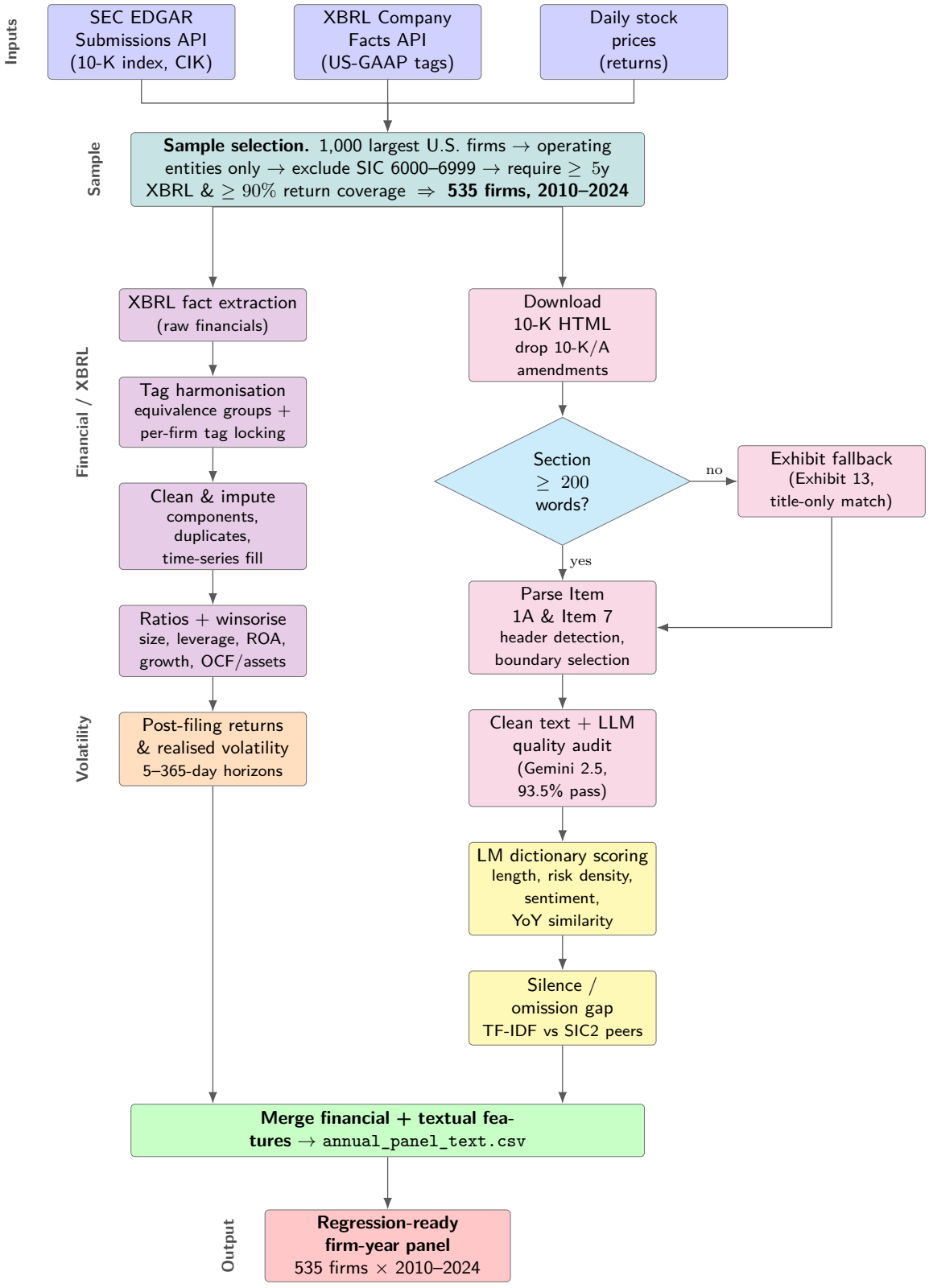


Figure 2.1: Operational pipeline for the complete data preparation. The workflow runs top-to-bottom from the raw SEC sources through sample selection, then splits into a financial (XBRL) branch and a textual (10-K narrative) branch that are merged into the final regression-ready firm-year panel. Own processing.

2.1 Sample selection

The starting list is the first 1,000 rows of the SEC file `company_tickers.json`, which maps tickers to Central Index Keys (CIK). The SEC orders the file approximately by market capitalisation but publishes no values or date with it; the version used was stored in the project repository in October 2026, so the ranking reflects firm size at that time, not in 2010. Three filters follow (Table A.1 in Appendix A). First, only entities that the SEC classifies as operating companies are kept, which removes exchange-traded funds, closed-end funds and shells (1,000 to 753). Second, firms with SIC codes 6000–6999 (finance, insurance and real estate) are excluded, as is standard, because their balance sheets and ratios are not comparable (753 to 600). Third, a firm needs at least five fiscal years of 10-K XBRL data (to 550 with today's metadata; the frozen list used for the panel contains 548 firms). Five years leave at least four years in the panel after the first year is used for asset growth, which gives each firm some time-series variation; the results do not depend on this threshold among the firms retained (Section 5.2).

Earlier versions of this panel also dropped firms with valid post-filing volatility for fewer than 90% of their filings. That filter used prices *after* each filing, that is, it selected on the outcome period, and it has been removed. Instead, a filing enters the regressions only if the stock has at least 200 daily closing prices in the 252 trading days *before* the filing date; this removes thinly traded or newly listed stocks using information available at the filing date and affects 1.3% of filings.

Sample size and look-ahead. Large firms have complete machine-readable filing histories, are liquid, and their filings are consistently formatted, which keeps measurement error in both branches low. The cost is that the list is drawn from today's large firms: firms that failed, were acquired or shrank during 2010–2024 are missing, and the panel grows from 152 firm-years in fiscal year 2010 to over 500 from 2020. The results therefore describe large surviving U.S. firms (Section 3.6). The 30-day estimation sample has 5,782 firm-years of 512 firms. By comparison, Campbell et al. (2014) study risk-factor disclosures of a broad cross-section of U.S. firms and Loughran and McDonald (2011) use about 50,000 10-K filings; their aims require broad coverage, whereas the within-industry-year comparisons here rely on a balanced set of large issuers.

2.2 Annual filings as the unit of observation

We use annual 10-K filings rather than quarterly 10-Q reports. A quarterly panel would have roughly four times as many observations, but two problems appeared in small pilot tests (Tables 2.1 and 2.2); the pilots are illustrations on a handful of firms, not representative estimates.

Financial data in the fourth quarter. 10-Q filings cover only the first three quarters, so a quarterly panel must reconstruct the fourth quarter as the annual figure minus the three quarters. In a pilot of 30 non-financial firms (Table 2.1), all three quarterly operating-cash-flow facts were available in only 7% of firm-years, because most firms report cumulative year-to-date cash flows. Operating cash flow enters the control vector, so a quarterly panel would lose it at the source. For net income and operating income the implied and the explicitly reported fourth quarter differ by more than 1% of the annual figure in about 10 to 14% of comparable firm-years. The residual absorbs year-end audit adjustments that are booked in the fourth quarter, whereas a later filing may report a recast figure, so the two represent different versions of the same accounting history.

Table 2.1: Q4 Reliability: Availability and Consistency of Fourth-Quarter XBRL Facts

Metric	Firm-Years	Q1 – Q3 (%)	Explicit Q4 (%)	Comparable	Match (%)
Net Income	451	80.9	43.0	172	89.5
Operating Income	391	79.0	22.0	72	86.1
Operating Cash Flow	462	7.4	3.2	0	—

Note: "Q1 – Q3" indicates the share of fiscal years where all three quarterly facts (Q1, Q2, Q3) exist in XBRL. "Explicit Q4" indicates the share with a standalone fourth-quarter fact (period ≈90 days ending at fiscal year-end). "Comparable" is the count of firm-years for which both an explicit Q4 fact and the implied residual are available. "Match" reports the share of those comparable observations where the two values agree within 1% of the annual figure. Sample: 30 non-financial firms, all available fiscal years.

Risk factors in quarterly reports. Form 10-Q (Part II, Item 1A) requires only *material changes* to the risk factors disclosed in the latest 10-K, whose content is governed by Item 105 of Regulation S-K (17 CFR 229.105). In a pilot of the six most recent 10-Q filings of 15 large firms, 52% contained a substantive risk-factor section and the rest referred back to the annual report (Table 2.2; Table A.7 lists the firms). Text variables measured on the full risk-factor section therefore cannot be built quarterly.

Annual filings contain audited full-year accounting data and complete risk-factor and MD&A sections, and they are the unit of analysis in the 10-K textual-analysis literature (Li, 2008; Loughran & McDonald, 2011). Matching the observation interval to the reporting frequency of the documents also means that changes in the text variables correspond to actual revisions of the annual disclosure.

Table 2.2: Textual Data Availability: 10-K vs. 10-Q Filings

	10-K	10-Q
Filings tested	44	90
MD&A (Management's Discussion and Analysis)		
Section found (%)	93	100
Median word count	20,494	5,806
Risk Factors		
Substantive section found (%)	100	52
Boilerplate / reference only (%)	0	48
Median word count (substantive)	29,902	9,144

Note: "Found" requires >200 words extracted by regex-based section parser. "Substantive" for 10-Q excludes filings containing only a reference to the annual 10-K (e.g., "no material changes from those disclosed in our Annual Report"). MD&A corresponds to Item 7 in 10-K and Part I Item 2 in 10-Q; Risk Factors corresponds to Item 1A in 10-K and Part II Item 1A in 10-Q. Sample: 15 firms, most recent filings.

2.3 Financial data from XBRL

Point-in-time values. For each firm and fiscal period the panel takes the values, the filing date and the accession number of the *first* 10-K that reports facts for that period end, filed within 180 days of it. Restated figures published later (in a 10-K/A or as comparatives in the next year's 10-K) are not mixed in, because they were not public when the post-filing window starts. A 10-K/A is used only when no original 10-K reports the period. All variables of a firm-year therefore come from one filing, identified by its accession number, which also identifies the filing whose text is used. The fiscal year is the calendar year in which the period ends (periods ending on 1–7 January count to the previous year, for firms with 52/53-week years).

Fiscal years and calendar time. Firms do not share a fiscal calendar, so the 10-Ks of one fiscal year are filed at different calendar dates (Table A.3 in Appendix A). 72.6% of the firm-years end in December, and 145 of the 512 firms in the estimation sample have a different fiscal-year end, from January to November. The median lag between period end and filing is 53 days (10th to 90th percentile: 38 to 59 days), so most filings fall in February, but the filings of one fiscal year span a median of 390 calendar days and fall in 169 distinct calendar months. Four choices keep this heterogeneity from distorting the analysis. First, the fiscal year is assigned from the period end itself, not from the fiscal-year end month that the SEC reports today, which is unreliable for firms with 52/53-week years and had shifted the labels of about 20 firms by one year in an earlier version of the panel. Second, every timing variable is anchored on the firm's own filing date, not on a common calendar date: the outcome window, the pre-filing windows and the lagged volatility of the previous filing all start at that

firm's day 0 (Section 3.1), and the accounting variables and the text come from the filing that opens the window. Third, the previous fiscal year is matched only when it is exactly one year earlier, so that a change of the fiscal-year end never produces a lag of unequal length. Fourth, because a fiscal-year effect cannot remove calendar-time shocks when the filings of one fiscal year are spread over more than a year, the robustness checks replace it by the calendar month of the filing and restrict the sample to firms with a December year end (Section 5.2).

Tag choices. One accounting concept can be reported under several US-GAAP elements, so the elements must be mapped explicitly (Chychyla & Kogan, 2015). Two kinds of alternatives are handled differently.

- *Equivalent tags* measure the same quantity and are used interchangeably. Net income is measured on a consolidated basis: `ProfitLoss` (including non-controlling interests, NCI) where a firm reports it, and `NetIncomeLoss` otherwise, which under US GAAP is then the same consolidated figure. This makes ROA consistent with total assets and total liabilities, which also include NCI. The two operating-cash-flow tags (total and continuing operations) are also treated as equivalent.
- *Non-equivalent tags*, such as stockholders' equity with or without NCI, or income from continuing operations versus net income, are *locked* per firm: each firm keeps the tag it reports in most years, and values under other tags are set to missing. The choice uses the firm's whole history, a mild look-ahead in which tag is used, not in the values.

Table A.4 in Appendix A reports the tag choices, the values lost to locking (637 of 7,900 equity values, 38 net-income values, no cash-flow values) and the differences between paired tags in firm-years that report both. Where both net-income tags are present (3,838 firm-years), the median absolute difference is 0.043% of total assets, but the 99th percentile is 3.56% and the maximum 37.4%; in 303 firm-years (7.9%) the difference exceeds 1% of total assets, which we call material. These differences are the NCI share of income, which the consolidated definition now includes consistently. For the two operating-cash-flow tags (417 firm-years) the difference exceeds 1% of assets in 50 firm-years (12.0%), with a median of 0.035%.

Missing balance-sheet values. If total liabilities are missing, they are computed as total assets minus stockholders' equity (2,171 firm-years). If total assets or liabilities are still missing, the value of the immediately preceding fiscal year is carried forward (1 and 32 firm-years); values are never filled from later years. Each imputation is flagged, and Section 5.2 re-estimates the models without imputed values.

Market capitalisation. Shares outstanding are taken from the same 10-K as the other variables, matched by accession number: the cover-page count (`dei:EntityCommonStockSharesOutstanding`, summed over share classes), else the balance-sheet count, else the fiscal-year weighted-average basic count. The last source is needed for multi-class firms whose cover-page counts are re-

ported per class and are not available in the Company Facts API (445 firm-years); a robustness check drops these firm-years. Shares are multiplied by the as-traded closing price on the last trading day before the filing, which is rebuilt from the split-adjusted price and the split history.

2.4 Text extraction from 10-K filings

The study uses Item 1A (Risk Factors) and Item 7 (Management's Discussion and Analysis, MD&A) of the 10-K.

2.4.1 Source of the text

For each original 10-K in the panel the primary HTML document was downloaded from the EDGAR archive, using the document name listed in the EDGAR Submissions API.[1] Where the filing incorporates a section by reference to an exhibit, the exhibit was fetched from the same filing (Section 2.4.2). All requests carried the declared user-agent header that the SEC requires and stayed below its limit of ten requests per second. The extraction was run once, in October 2026; the stored texts were not re-extracted for this version, and each stored section is linked to the accession number of the filing it came from. In seven firm-years the stored text comes from a 10-K/A whereas the panel now uses the original 10-K; these firm-years are excluded from all text regressions.

2.4.2 Section identification and exhibit fallback

The text of a 10-K is embedded in HTML without standard tagging, mixed with table-of-contents entries, running headers and cross-references, so a plain search for "Item 1A" typically finds the table of contents. The parser works on the document structure before the text is flattened:

1. **Candidate detection.** Elements (paragraphs, divisions, table cells, headings) whose text matches an "Item X" pattern are kept as candidate headers if they have at most 25 words, carry a heading cue (bold font, all capitals, or a font size of at least 12 pt), and contain no hyperlink (table-of-contents links).
2. **In-place tagging.** Each candidate is marked inside the HTML tree, and only then is the document converted to text, so header positions survive the conversion.
3. **Candidate selection.** A candidate for Item 1A must have the word "risk" within 300 characters, and one for Item 7 "management", "discussion" or "MD&A". Candidates with at least three other item headers within 300 characters (a table of contents) and

[1] `https://data.sec.gov/submissions/CIK{cik}.json`; see https://www.sec.gov/search-filings/edgar-application-programming-interfaces.

candidates preceded by “see”, “refer to” or similar phrases are discarded when alternatives exist. Among the rest the parser takes the candidate followed by the longest section.

4. **Section bounding.** The section ends at the next item header: Item 1B, 1C or 2 for Item 1A, and Item 7A, 8, 9, 9A or 9B for Item 7. Several terminators are needed because some items are optional (Item 1B applies only to some filers, and Item 1C exists only for fiscal years ending on or after 15 December 2023). A section needs at least 200 words.

Exhibit fallback. Some filers, especially early in the sample, incorporate the MD&A by reference to the annual report to shareholders, attached as an exhibit (usually Exhibit 13). If no section of at least 200 words is found and the text under the header (fewer than 500 words) refers to an exhibit or the annual report, the parser applies the same procedure to the referenced exhibit, using title patterns (“Risk Factors”, “Management’s Discussion and Analysis”) because exhibits have no item numbers. The extraction run did not record which sections came from an exhibit, and the share cannot be measured without downloading the filings again. We expect it to be small and concentrated in Item 7 of early years: Item 1A must appear in the body of the 10-K itself, and incorporation by reference to the annual report was mainly used for MD&A.

2.4.3 Post-processing and tokenisation

During the HTML conversion, scripts, style sheets and the hidden inline-XBRL header are removed, and tables are removed if they are numerical: at least 30% of their non-empty cells (with at least four cells and two rows) are bare numbers or currency amounts, or more than 40% of their characters are digits or number symbols. Other tables, which in many filings are used for page layout, keep their text. After the section has been bounded, the parser removes running “Table of Contents” headers, page footers such as “2019 Form 10-K 59”, standalone page numbers, residual numeric table rows (lines of at least four tokens of which more than 40% are numbers), and short lines that repeat at least four times (running headers), and replaces non-breaking spaces. Footnote markers and text of layout tables are not removed.

For the dictionary measures the text is lowercased and split into tokens of at least two letters (digits, hyphens and apostrophes split tokens); a section with fewer than 100 tokens is treated as missing. The dictionary is the Loughran–McDonald Master Dictionary, 1993–2025 version (Loughran & McDonald, 2011), with 297 uncertainty and 903 litigious words; words that the dictionary marks as removed are not counted. The two lists share no word.

2.4.4 Coverage

Of the 6,757 firm-years of the panel, 96.3% have an Item 1A of at least 100 tokens and 94.9% an Item 7 (Table A.2 in Appendix A). Coverage is lowest in fiscal year 2010 (92.8%) and is 94.7% in 2024; the lower 2024 figure comes from a correction of the fiscal-year labels of about 15 firms with 52/53-week years, whose fiscal-year-2024 filing had not been extracted.

Extraction audits. Two large language models (LLMs) audited a stratified random sample of stored sections. The sample has five three-year strata (2010–2012 to 2022–2024) with 20 sections of each type per stratum, 200 sections in total; Item 1A and Item 7 were drawn independently, so the 200 sections come from 197 filings. Each section was rated on five criteria (correct section, start boundary, end boundary, content quality, completeness) as Pass, Minor issue or Fail. The first audit, by Google Gemini 2.5 Flash, saw the full extracted section; a second, independent audit by Claude Opus 5.5 saw the first and last 1,200 characters, a middle excerpt and the word count, without the first audit's ratings. Neither saw the rest of the filing, so boundary judgements rest on the extracted text alone. LLMs perform well on text-classification tasks compared with crowd workers (Gilardi et al., 2023), but their judgements are not a substitute for an expert reading of the filings. The author therefore also checked 51 of the sections by hand against the original filings on EDGAR, without seeing the LLM ratings: 50 drawn at random within the same strata and the one further section flagged by an LLM.

Under the first audit, 187 of 200 sections (93.5%, Wilson 95% interval 89.2–96.2%) were acceptable on all five criteria (Pass or Minor) and 179 (89.5%) were rated Pass on all five. The 13 sections with a Fail include five with no extracted text; by stratum, five fall in 2010–2012, three in 2013–2015, two each in 2016–2018 and 2019–2021, and one in 2022–2024. Under the second audit 190 sections (95.0%, 91.0–97.3%) were acceptable; the two audits agree on 96.5% of sections ($\kappa = 0.68$) and both reject the same eight (Table B.1 in Appendix B). In the manual check every one of the 51 sections was the correct section; four had a boundary error (two started inside Item 1, one ran a few lines into the financial statements and one ran through Item 8), three of them among the 50 random sections, an error rate of 6.0% (Wilson 95% interval 2.1–16.2%). The manual verdict on the three criteria matches the first audit on all 51 sections and the second on 50. Errors are more frequent in early filings, whose structure is less standardised.

Relation to other approaches. Changes in the structure of 10-K filings over time make large-sample extraction difficult (Dyer et al., 2017). Sections tagged in XBRL (for example `us-gaap:RiskFactorsTextBlock`) are incomplete and inconsistent in early years, so a parser that works on the HTML lets the same method be applied to the whole 2010–2024 period.

3. Methodology

This chapter sets out the regression specifications for the three hypotheses. All tests regress post-filing realised volatility on characteristics of the Item 1A risk-factor section, a common vector of controls known when the 10-K is filed, and fixed effects. Section 3.1 defines the dependent variable and the timing of all variables; Section 3.2 lists the controls, which are the same in every specification so that the text coefficients can be compared across hypotheses. Sections 3.3 and 3.4 describe the joint test of the volume channel (H1) and the composition channel (H2), which are estimated together because the density measure for H2 has section length in its denominator. Section 3.5 introduces the peer-relative omission gap for H3 and the decision rule used to evaluate it, and Section 3.6 covers estimation and inference.

3.1 Outcome variable and timing

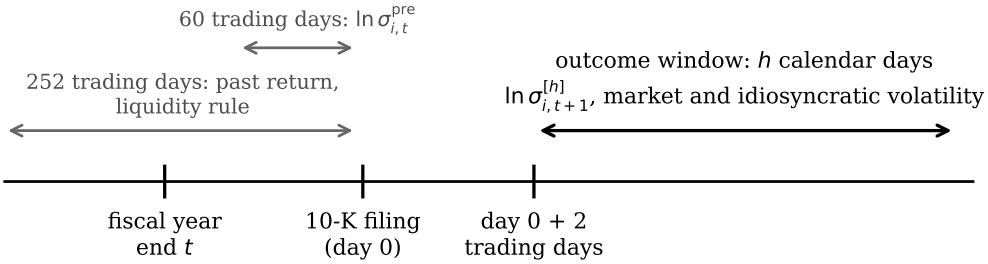
The dependent variable is the natural logarithm of annualised post-filing stock-return volatility measured over a window of h calendar days:

$$\ln \sigma^{[h]}_{i,t+1} \;=\; \ln\!\left(\sqrt{252}\cdot \mathrm{sd}\big(r_{i,d}\big)_{d\in W^{[h]}_{i,t}}\right), \tag{3.1}$$

where $r_{i,d} = \ln(P_{i,d}/P_{i,d-1})$ is the daily log return of firm i computed from dividend- and split-adjusted closing prices, and $W^{[h]}_{i,t}$ is the post-filing window that belongs to the 10-K for fiscal year t. The subscript $t+1$ marks that the window lies after the filing; it is not the next fiscal year. Day 0 is the first trading day on or after the filing date. The window begins two trading days after day 0, so the market reaction on the filing day and the next day is left out, and it then runs for h calendar days. Only returns between two prices inside the window are used. A window needs a minimum number of daily returns (3 for $h = 5$, 5 for $h = 10$, 14 for $h = 30$, 44, 89 and 199 for the longer windows); with fewer, or a zero standard deviation, the volatility is missing. A window whose nominal end lies after the last trading day of the price data (30 October 2026) is also missing rather than truncated; with this cut-off every fiscal-year 2024 filing has a complete 365-day window (Section 2.1).

Figure 3.1 shows the timing. Every control and every text variable is measured from information that is public at day 0: the accounting variables and the text come from the same 10-K, and the price-based controls use windows that end on the last trading day before the filing date.

A coefficient β on a regressor x implies a change of $100(e^{\beta\Delta x} - 1)\%$ in volatility for a change Δx. The headline horizon is $h = 30$ days. The short window keeps the outcome close to the information release whose content is measured, so that later information events weigh less; the 10-K literature that relates filing text to subsequent volatility uses post-filing windows measured in days to months as well as a full year (Campbell et al., 2014; Kogan et al., 2009;



Accounting controls: fiscal year t (reported in this 10-K); text variables: Item 1A of this 10-K. Not to scale.

Figure 3.1: Timing of the variables for the 10-K of fiscal year t (not to scale). The outcome window starts two trading days after day 0 and lasts h calendar days; pre-filing volatility uses the 60 trading days, and the past return and the liquidity rule the 252 trading days, that end on the last trading day before the filing date.

Loughran & McDonald, 2011). The 365-day window is reported alongside the headline in every main table, and $h \in \{5, 10, 90, 180\}$ are reported in the robustness chapter.

3.2 Control variables

Every regression for H1, H2 and H3 includes the same vector $\mathbf{X}_{i,t}$ of nine controls, all known at day 0:

- pre-filing volatility $\ln \sigma_{i,t}^{\text{pre}}$: the log of the annualised standard deviation of the 60 daily returns ending on the last trading day before the filing date;
- firm size ln(Market cap): the as-traded closing price on the last trading day before filing times the shares outstanding reported in the same 10-K;
- book-to-market: book equity attributable to the parent over market capitalisation;
- the past 12-month log return, over the 252 trading days before filing;
- leverage (total liabilities over total assets), return on assets (consolidated net income over total assets), asset growth, the current ratio, and operating cash flow over total assets.

Pre-filing volatility is the single most important control: volatility is persistent, and a disclosure written in a volatile period may simply reflect that period. Size, book-to-market and past returns are the standard cross-sectional determinants of return volatility, and the accounting ratios follow the risk-disclosure literature (Campbell et al., 2014; Kogan et al., 2009). Operating cash flow over assets is a control like the others, not a robustness check.

The earlier version of this model used the volatility of the same-length window after the previous year's 10-K as the dynamic control. Once pre-filing volatility is included, that lag adds nothing (its coefficient is $+0.031$ with $t = +0.69$ in the industry fixed-effects model; Table 5.4 reports the firm fixed-effects version), so it is reported only as a robustness check. Each firm's first fiscal year is not in the sample, because asset growth needs the previous year's total assets. Financial ratios, book-to-market and the past return are winsorised at the 1st and 99th percentiles of each regression sample; market capitalisation and pre-filing volatility enter in logs and are not winsorised.

3.3 H1 — Item 1A length and disclosure volume

Following Campbell et al. (2014), disclosure volume is measured as the logarithm of the Item 1A word count. Campbell et al. find that firms with longer risk-factor sections have higher subsequent market-based risk measures, consistent with the volume of risk discussion carrying information about firm risk. The volume variable is

$$\text{Words}_{i,t}^{1A} \;=\; \text{number of tokens in Item 1A of firm } i\text{'s fiscal-year-}t \text{ 10-K}, \tag{3.2}$$

where a token is a run of at least two letters in the lowercased text (Section 2.4.3). The cleaning removes running page headers, page numbers and numeric table rows but keeps the text of layout tables and footnote markers, so the count is close to, but not exactly, the narrative content. The median Item 1A in the estimation sample has 8,700 tokens (Table 4.1).

H1. *Holding the composition of Item 1A and the controls fixed, a longer Item 1A is associated with higher post-filing volatility*: $\beta^{\text{len}} > 0$ in

$$\ln \sigma^{[h]}_{i,t+1} = \alpha + \boldsymbol{\gamma}'\mathbf{X}_{i,t} + \beta^{\text{len}} \ln(\text{Words}^{1A}_{i,t}) + \beta^{\text{dens}} \text{RiskDensity}_{i,t} + \mu_{j(i,t)} + \tau_t + \varepsilon_{i,t}, \quad (3.3)$$

where μ_j is a fixed effect for the two-digit SIC industry of the firm at the filing date, τ_t a fiscal-year fixed effect, and RiskDensity is defined in Section 3.4. Length and density enter together because the density has length in its denominator.

3.4 H2 — Risk-word density

H2 concerns the composition of Item 1A at a given length: the share of words from the uncertainty and litigious dictionaries of Loughran and McDonald (2011), which were built for financial reporting,

$$\text{RiskDensity}_{i,t} = \frac{n^{\text{unc}}_{i,t} + n^{\text{lit}}_{i,t}}{\text{Words}^{1A}_{i,t}}, \quad (3.4)$$

where n^{unc} and n^{lit} count the tokens of Item 1A that belong to each list. The two lists share no word in the dictionary version used (Section 2.4.3), so no token is counted twice.

The sign of the association is not determined by theory, and H2 is therefore stated two-sided. Two channels pull in opposite directions.

- *Risk-content channel.* Uncertainty and legal-exposure words name risks. At a given length, a section dense in such words discusses more risk, and if the discussion is informative about firm risk (Campbell et al., 2014; Kravet & Muslu, 2013), density should be associated with *higher* volatility.
- *Boilerplate channel.* Much of this vocabulary is standard cautionary language that firms repeat across years and that applies to any firm (Cazier et al., 2021; Dyer et al., 2017). At a given length, a section dense in standard risk terms leaves less room for firm-specific discussion, which is the kind of risk disclosure that investors find informative (Hope et al., 2016). Under this channel density is associated with *lower* volatility.

H2. *Holding length and the controls fixed, RiskDensity is associated with post-filing volatility*: $\beta^{\text{dens}} \neq 0$ in equation (3.3). The sign identifies the dominating channel. Density is a proxy for composition, not a direct measure of specificity in the sense of Hope et al. (2016): words such as "litigation" or "investigation" can describe very firm-specific circumstances.

Sub-list decomposition. The second stage replaces RiskDensity by its two components, $\text{UncDensity} = n^{\text{unc}}/\text{Words}^{1A}$ and $\text{LitDensity} = n^{\text{lit}}/\text{Words}^{1A}$:

$$\ln \sigma^{[h]}_{i,t+1} = \alpha + \boldsymbol{\gamma}'\mathbf{X}_{i,t} + \beta^{\text{len}} \ln(\text{Words}^{1A}_{i,t}) + \beta^{\text{unc}} \text{UncDensity}_{i,t} + \beta^{\text{lit}} \text{LitDensity}_{i,t} + \mu_{j(i,t)} + \tau_t + \varepsilon_{i,t}. \quad (3.5)$$

The uncertainty list covers ambiguity and incomplete knowledge of outcomes; the litigious list covers legal exposure, regulation, investigations and disputes (Loughran & McDonald, 2011). A Wald test of $\beta^{\text{unc}} = \beta^{\text{lit}}$ asks whether the two behave differently.

Ratio regressors. Densities are ratios with length in the denominator, and length is also a regressor. A regression on ratios can produce associations that come from the denominator alone (Kronmal, 1993). Two features limit this. First, length is in the model, so the density coefficient is estimated at a given length. Second, a count specification replaces the densities by $\ln(1 + n^{\text{unc}})$ and $\ln(1 + n^{\text{lit}})$ next to $\ln(\text{Words}^{1A})$, which uses no ratio at all (Section 5.2).

3.5 H3 — Peer-relative omission gap

H3 asks whether what a firm *leaves out* of Item 1A relative to its industry peers is associated with subsequent volatility, beyond length and density. The *omission gap* is the measure; *strategic silence* is one interpretation of it, namely that a firm stays quiet on risks its peers disclose. Non-strategic explanations predict the same correlation: a firm may not face the risk, may describe it in different words, or may have a shorter section for reasons unrelated to risk.

Construction. For each fiscal year t, a TF–IDF model is fitted on the Item 1A texts filed before the first fiscal-year-t filing in the panel, so that vocabulary and weights are known at every focal filing date (fiscal year 2010 has no earlier texts and therefore no omission gap). The settings are those of `scikit-learn`: lowercased text, tokens of at least two letters or digits, the library's English stop-word list removed, single words only, terms in at least five and at most 95% of the training documents, raw term counts, smoothed inverse document frequency, and rows normalised to unit length. Let $\mathbf{x}_{i,t}$ be the TF–IDF row of firm i's fiscal-year-t Item 1A. The peers of a focal filing are the other panel firms in the same two-digit SIC industry (at the filing date); for each peer the most recent Item 1A of fiscal year t or $t-1$ filed strictly before the focal filing is used, so no later information enters. With $P_{i,t}$ this peer set,

$$\mathbf{p}_{i,t} = \sum_{k \in P_{i,t}} \mathbf{x}_k, \qquad \text{OMISSION}^{1A}_{i,t} = 1 - \frac{\sum_{w: x_{i,t,w} > 0} p_{i,t,w}}{\sum_w p_{i,t,w}}, \quad (3.6)$$

the share of peer TF–IDF mass on terms the focal firm does not use. It lies in [0,1]. Filings with fewer than five peers get no value.

Length and the sources of variation. Omission is mechanically related to length, because a longer section uses more of the peer vocabulary; in the H3 sample the correlation with $\ln(\text{Words}^{1A})$ is -0.915 (Table 5.7). Three variants address this and the question of where its variation comes from:

- a *length-invariant* omission gap: for each filing, 20 random contiguous blocks of 3,000 tokens are drawn from its Item 1A (fixed seed), the gap is computed with each block in place of the full text, and the 20 values are averaged; sections shorter than 3,000 tokens use the whole text;
- a *pooled-vocabulary* gap, with one TF–IDF model fitted on all years, which removes changes in the vocabulary over time (it uses later documents and is a diagnostic only);
- a decomposition of the year-on-year change of the gap into the change of the firm's own text (peers held at $t-1$), the change of the peers' texts, and the change in peer composition.

Specification. The H3 regression adds the omission gap to equation (3.3):

$$\ln \sigma^{[h]}_{i,t+1} = \alpha + \boldsymbol{\gamma}'\mathbf{X}_{i,t} + \beta^{\text{len}} \ln(\text{Words}^{1A}_{i,t}) + \beta^{\text{dens}} \text{RiskDensity}_{i,t} + \beta^{\text{om}} \text{Omission}^{1A}_{i,t} + \mu + \tau_t + \varepsilon_{i,t}, \quad (3.7)$$

estimated with industry fixed effects ($\mu = \mu_j$) and with firm fixed effects ($\mu = \mu_i$). The controls are the common vector of Section 3.2, so the H3 model differs from the H1/H2 model only by the omission gap.

H3 and the decision rule. *Holding length, density and the controls fixed, a larger omission gap is associated with higher post-filing volatility*: $\beta^{\text{om}} > 0$. The evidence is judged by a fixed rule, recorded in the project log before the models were re-estimated:

1. The family of tests is $h \in \{5, 10, 30, 90, 180\}$ days under both fixed-effect structures (ten two-sided tests); p-values are adjusted with the Holm step-down procedure (Holm, 1979) at $\alpha = 0.05$.
2. The primary design is $h = 30$ days. Under a given fixed-effect structure there is *positive evidence* if $\beta^{\text{om}} > 0$ with Holm-adjusted $p < 0.05$ *and* the length-invariant gap has a positive coefficient with $p < 0.05$ in the same specification; *negative evidence* is the mirror image.
3. H3 is *supported* with positive evidence under both structures, *partially supported* with positive evidence under one and no negative evidence under the other, *contradicted* with only negative evidence, and *not supported* otherwise. The two fixed-effect structures are treated symmetrically.

The placebo test, the decomposition and the other robustness checks are reported but do not change the verdict.

3.6 Estimation and inference

All regressions are estimated by ordinary least squares with the fixed effects absorbed (`linearmodels`). With industry and year fixed effects, the coefficients use variation *between* firms within an industry-year cell. Firm fixed effects are used as a robustness check (Section 5.3); they use only year-to-year variation *within* a firm. The two answer different questions, and a coefficient estimated under industry fixed effects is not evidence of a within-firm relation. The analysis is predictive, not causal: the coefficients measure conditional associations between the disclosure and subsequent volatility, as in Campbell et al. (2014).

Fiscal-year effects. The year fixed effects absorb shocks common to all firms of a fiscal year. Because fiscal years end in different months, the filings of one fiscal year are spread over more than a year of calendar time (Section 2.3), so a fiscal-year effect does not remove calendar-time market shocks: the volatility spike of spring 2020, for example, falls in the post-filing windows of firms with different fiscal-year labels and in only part of the windows within one label. Four robustness checks address this: the market (SPY) volatility over the same window as a control, filing-month $\times$ fiscal-year fixed effects, fixed effects for the calendar month of the filing in place of the fiscal-year effects, and the sample of firms with a December fiscal-year end, for which fiscal and calendar time coincide. The idiosyncratic volatility (the residual standard deviation of a market-model regression inside the window) is used as a further dependent variable.

Anticipated volatility. Item 1A is written when management already knows current conditions, so expected volatility could shape the wording. Pre-filing volatility, which is public at filing, captures the part of this that is visible in prices. It does not rule out that management anticipates future volatility beyond it, so all results are read as associations.

Standard errors. Standard errors are clustered two-way by firm and fiscal year (Cameron et al., 2011; Petersen, 2009). Firm clusters allow for persistent firm-specific shocks that make a firm's residuals correlated over years; year clusters allow for shocks common to all firms in a fiscal year, such as the 2020 market turmoil. With 15 fiscal years the year dimension has few clusters, and cluster-robust tests can then over-reject (Cameron & Miller, 2015; Cameron et al., 2008). For the text coefficients we therefore also report p-values from a wild cluster bootstrap by fiscal year with Webb weights and 9,999 draws (Webb, 2023), and from one-way firm clustering.

Dynamics and Nickell bias. With firm fixed effects and at most 15 years per firm, a lagged dependent variable is correlated with the within-transformed error, and its coefficient is biased towards zero by an amount of order $1/T$ (Nickell, 1981); the bias can carry over to the other coefficients. The firm fixed-effects models therefore use pre-filing volatility, which is measured

before the window and is not a lag of the dependent variable, as the dynamic control, and report the version with the lagged dependent variable only for comparison. A related concern, spurious regression from integrated series, is less relevant with $T \leq 15$ and large N, but Section 5.5 reports panel unit-root tests designed for this case.

Outliers. Accounting ratios, book-to-market and the past return are winsorised as described in Section 3.2. The text variables are not winsorised: $\ln(\text{Words}^{1A})$ ranges from 7.41 to 10.48 between the 1st and 99th percentiles, and RiskDensity has a maximum of 0.0772 (Table 4.1).

Sample. The 30-day estimation sample has 5,782 firm-years of 512 firms; the same firm-years enter the 365-day models. Table A.1 in Appendix A traces the sample from the SEC ticker list to this sample.

Limitation: survivorship and look-ahead. The universe is the first 1,000 rows of the SEC ticker list as stored in October 2026, which is ordered by approximately current size. Firms that were delisted, acquired or failed during 2010–2024 are absent, and firms that are large today are over-represented in early years: the panel grows from 152 firm-years in fiscal year 2010 to over 500 from 2020 (Table A.2). Firms in distress are likely to be riskier, so the results describe large surviving firms. The sample rules themselves look only backwards: the liquidity requirement (Section 2.1) uses prices before the filing.

4. Results

This chapter reports the descriptive statistics and the tests of H1, H2 and H3 at the headline 30-day horizon.

4.1 Descriptive statistics

Table 4.1 summarises the 30-day estimation sample of 5,782 firm-years. The median annualised 30-day post-filing volatility is 0.243 (mean 0.317). The median Item 1A has 8,700 tokens, with a long right tail (99th percentile 35,579). The mean RiskDensity is 0.0511, so about one token in twenty is an uncertainty or litigious word; the uncertainty sub-list (mean density 0.0346) is about twice as dense as the litigious one (0.0165).

Table 4.1: Summary statistics (30-day estimation sample)

	Mean	SD	P1	P25	Median	P75	P99
$\sigma^{[30]}_{i,t+1}$	0.317	0.269	0.089	0.176	0.243	0.346	1.465
$\ln \sigma^{[30]}_{i,t+1}$	-1.347	0.579	-2.423	-1.736	-1.416	-1.063	0.382
$\ln \sigma^{\text{pre}}_{i,t}$	-1.294	0.433	-2.181	-1.597	-1.335	-1.042	-0.103
ln(Market cap)	23.591	1.436	20.252	22.737	23.548	24.451	26.983
Leverage	0.612	0.219	0.128	0.472	0.610	0.737	1.298
ROA	0.064	0.089	-0.319	0.028	0.063	0.107	0.294
Asset growth	0.117	0.247	-0.241	0.003	0.060	0.144	1.540
Book-to-market	0.299	0.255	-0.122	0.116	0.243	0.423	1.265
Past 12-month return	0.135	0.285	-0.753	-0.020	0.144	0.289	1.003
Current ratio	1.955	1.477	0.379	1.052	1.473	2.346	8.869
OCF/assets	0.112	0.079	-0.196	0.069	0.106	0.153	0.324
Words^{1A}	10,053	6,990	1,652	5,504	8,700	12,253	35,579
$\ln(\text{Words}^{1A})$	9.012	0.649	7.410	8.613	9.071	9.414	10.479
RiskDensity	0.0511	0.0065	0.0358	0.0471	0.0508	0.0552	0.0673
UncDensity	0.0346	0.0049	0.0225	0.0317	0.0346	0.0374	0.0472
LitDensity	0.0165	0.0046	0.0064	0.0134	0.0162	0.0194	0.0280
Observations	5,782						

Notes. $\sigma^{[30]}_{i,t+1}$: annualised standard deviation of daily log returns in the 30-calendar-day window starting two trading days after the 10-K filing day. σ^{pre}: the same over the 60 trading days before filing. Financial ratios, book-to-market and the past return are winsorised at the 1st/99th percentiles of this sample; the text variables are not winsorised. The maximum RiskDensity is 0.0772.

Figure 4.1 plots the median 30-day post-filing volatility by fiscal year with the inter-quartile range. Volatility peaks for fiscal year 2019, whose 10-Ks were filed in early 2020, just before and during the market turmoil of spring 2020. The fiscal-year fixed effects absorb such common shocks to the extent that they hit all filings of a fiscal year.

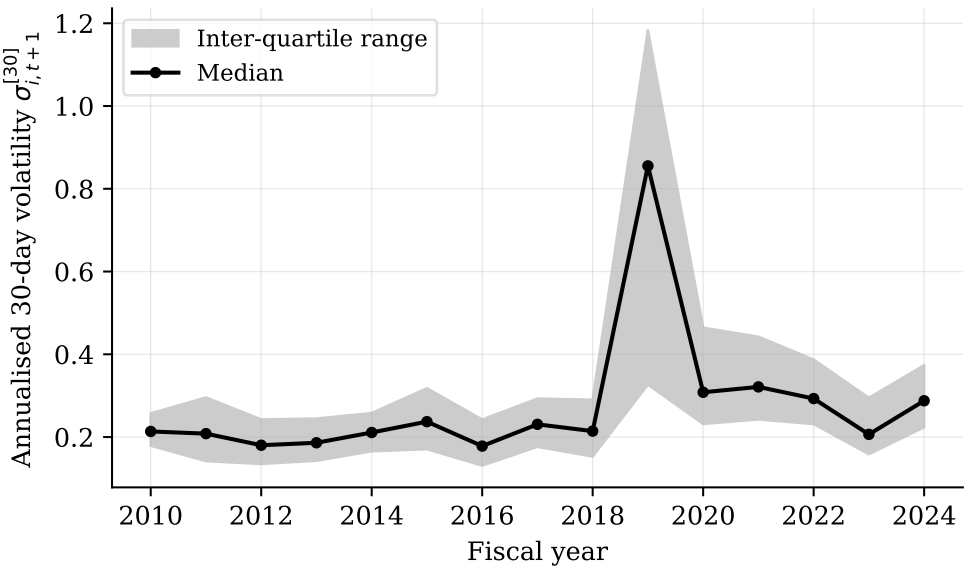


Figure 4.1: Median annualised 30-day post-filing volatility by fiscal year, with the inter-quartile range (shaded), 30-day estimation sample.

Table 4.2 reports pairwise correlations. Length and RiskDensity are negatively correlated ($r = -0.30$): longer sections have a lower share of dictionary risk words, mainly uncertainty words ($r = -0.44$). The two sub-densities are almost uncorrelated ($r = -0.07$). Pre-filing and post-filing volatility have a correlation of 0.53, which is why pre-filing volatility is the key control.

Table 4.2: Pearson correlations (30-day estimation sample)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(1) $\ln \sigma^{[30]}_{i,t+1}$													
(2) $\ln \sigma^{\text{pre}}_{i,t}$	0.53												
(3) ln(Market cap)	-0.17	-0.29											
(4) Leverage	-0.07	-0.12	0.11										
(5) ROA	-0.23	-0.30	0.25	-0.04									
(6) Asset growth	0.15	0.15	-0.04	-0.12	-0.03								
(7) Book-to-market	0.04	0.03	-0.25	-0.30	-0.25	-0.06							
(8) Past 12-month return	-0.08	-0.16	0.08	0.00	0.10	0.10	-0.28						
(9) Current ratio	0.18	0.26	-0.19	-0.50	-0.06	0.19	-0.10	0.04					
(10) OCF/assets	-0.16	-0.17	0.25	-0.00	0.76	-0.07	-0.32	0.15	-0.07				
(11) ln(Words^{1A})	0.31	0.36	-0.02	-0.04	-0.28	0.16	-0.00	-0.01	0.17	-0.19			
(12) RiskDensity	-0.11	-0.17	0.02	-0.01	0.06	-0.07	-0.04	0.02	-0.08	-0.02	-0.30		
(13) UncDensity	-0.10	-0.10	-0.14	-0.06	0.14	-0.05	-0.04	0.03	-0.00	0.08	-0.44	0.71	
(14) LitDensity	-0.06	-0.13	0.18	0.05	-0.08	-0.05	-0.02	-0.00	-0.10	-0.12	0.05	0.65	-0.07

Notes. N as in Table 4.1.

4.2 Length and risk-word density (H1 and H2)

Table 4.3 reports five nested specifications on the same sample: the controls only (1), length (2), density (3), length and density jointly (4), and the sub-list decomposition (5).

Table 4.3: Item 1A length, risk-word density and post-filing volatility (30-day window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\text{Words}^{1A})$		+0.061***		+0.053***	+0.058***
		(+5.40)		(+4.51)	(+4.54)
RiskDensity			−3.408***	−1.845*	
			(−3.33)	(−1.70)	
UncDensity					−0.545
					(−0.31)
LitDensity					−3.018***
					(−2.90)
$\ln \sigma_{i,t}^{\text{pre}}$	+0.614***	+0.588***	+0.603***	+0.585***	+0.584***
	(+10.88)	(+9.96)	(+10.27)	(+9.78)	(+9.85)
ln(Market cap)	−0.026***	−0.028***	−0.027***	−0.028***	−0.027***
	(−3.13)	(−3.35)	(−3.17)	(−3.35)	(−3.08)
Leverage	+0.038	+0.030	+0.030	+0.026	+0.027
	(+1.22)	(+0.96)	(+0.94)	(+0.84)	(+0.86)
ROA	−0.337***	−0.263***	−0.321***	−0.263***	−0.264***
	(−3.47)	(−2.71)	(−3.29)	(−2.73)	(−2.74)
Asset growth	+0.136**	+0.122**	+0.131**	+0.121**	+0.120**
	(+2.32)	(+2.11)	(+2.27)	(+2.10)	(+2.06)
Book-to-market	+0.021	+0.003	+0.015	+0.002	+0.003
	(+0.69)	(+0.11)	(+0.47)	(+0.06)	(+0.11)
Past 12-month return	+0.023	+0.015	+0.022	+0.015	+0.015
	(+0.51)	(+0.32)	(+0.48)	(+0.33)	(+0.32)
Current ratio	+0.022***	+0.018***	+0.020***	+0.017***	+0.017***
	(+4.49)	(+3.62)	(+4.02)	(+3.48)	(+3.47)
OCF/assets	−0.064	−0.096	−0.106	−0.114	−0.122
	(−0.38)	(−0.56)	(−0.61)	(−0.66)	(−0.72)
Industry and year FE	Yes	Yes	Yes	Yes	Yes
Observations	5,782	5,782	5,782	5,782	5,782
Firms	512	512	512	512	512
Within R^2	0.363	0.367	0.365	0.368	0.368
Δ within R^2 vs (1)	0.0000	0.0042	0.0018	0.0047	0.0049
Wald $\beta^{\text{unc}} = \beta^{\text{lit}}$, p					0.179

Notes. Dependent variable $\ln \sigma_{i,t+1}^{[30]}$. All columns use the same sample and the common control vector. Industry = two-digit SIC code at the filing date. t-statistics in parentheses use standard errors clustered two-way by firm and fiscal year. Within R^2: share of the variance left after the fixed effects that the regressors explain. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

H1. In the joint specification (column 4) a longer Item 1A is associated with higher post-filing volatility: $\hat{\beta}^{\text{len}} = +0.053$ ($t = +4.51$). The coefficient is similar in columns 2 and 5 and remains significant with a wild cluster bootstrap by year ($p = 0.001$; Table 5.3). **H1 is supported** in the industry fixed-effects design. Section 5.3 shows that the association does not hold within firms over time.

H2. Holding length fixed, the composite RiskDensity has a negative coefficient ($\hat{\beta}^{\text{dens}} = -1.845$, $t = -1.70$, $p = 0.090$; wild bootstrap $p = 0.092$). Without the length control (column 3) it is -3.408 ($t = -3.33$): length and density are negatively correlated, so part of the density coefficient in column 3 reflects length. Because H2 is two-sided, the composite test does not reject $\beta^{\text{dens}} = 0$ at the 5% level: **H2 is not supported at the 30-day horizon**. At longer horizons the composite coefficient is negative and significant at the 5% level from 90 days on ($t = -3.33$ at 365 days; Table 4.5).

Sub-lists. In column 5 the uncertainty density has no association with volatility ($\hat{\beta}^{\text{unc}} = -0.545$, $t = -0.31$), and the litigious density a negative one ($\hat{\beta}^{\text{lit}} = -3.018$, $t = -2.90$; wild bootstrap $p = 0.005$). The Wald test does not reject that the two coefficients are equal ($p = 0.179$), so the data do not establish that the two sub-lists behave differently; the negative sign of the composite comes mainly from the litigious words. The negative sign is in line with the boilerplate channel of Section 3.4 rather than the risk-content channel, but the design cannot tell which feature of litigious language drives it.

Magnitudes. Table 4.4 expresses the coefficients on a common basis. An increase of one pooled standard deviation in $\ln(\text{Words}^{1A})$ (0.65 log points) is associated with 3.8% higher 30-day volatility, and an increase of one pooled standard deviation in LitDensity with 1.4% lower volatility; for the composite RiskDensity the figure is -1.2%. Variation within a firm is much smaller (one within-firm standard deviation of length is 0.27 log points), so the same coefficients imply 1.6% and -0.6% for within-firm changes. At the median volatility of 0.243 a 3.8% change is about 0.9 percentage points of annualised volatility. The text variables add little explained variance: the within-R^2 rises from 0.363 with the controls alone to 0.368 (Table 4.3).

365-day horizon. Table 4.5 repeats the five columns with the 365-day window on the same firm-years. Length remains positive ($+0.084$, $t = +8.64$ in column 5), LitDensity negative (-4.709, $t = -5.38$) and UncDensity insignificant; the Wald test rejects equality of the two sub-list coefficients at this horizon ($p < 0.001$).

Table 4.4: Economic magnitude of the text coefficients (30-day window)

FE	Variable	SD pooled	SD within firm	% per pooled SD	% per within SD
Industry + year	$\ln(\text{Words}^{1A})$	0.6488	0.2748	+3.83	+1.60
Industry + year	UncDensity	0.0049	0.0022	-0.27	-0.12
Industry + year	LitDensity	0.0046	0.0020	-1.37	-0.61
Firm + year	$\ln(\text{Words}^{1A})$	0.6488	0.2748	+0.56	+0.23
Firm + year	UncDensity	0.0049	0.0022	-1.16	-0.52
Firm + year	LitDensity	0.0046	0.0020	-1.42	-0.63
Industry + year	RiskDensity (col. 4)	0.0065	0.0030	-1.19	-0.55
Firm + year	RiskDensity (col. 4)	0.0065	0.0030	-1.75	-0.81

Notes. Change in annualised 30-day volatility, $100\,(e^{\hat{\beta}\,\Delta x}-1)$, for an increase of one pooled standard deviation or one within-firm standard deviation (deviation from the firm mean) of the variable, using the coefficients of Table 4.3 (industry FE) and Table 5.4 (firm FE).

Table 4.5: Item 1A length, risk-word density and post-filing volatility (365-day window)

	(1)	(2)	(3)	(4)	(5)
$\ln(\text{Words}^{1A})$		+0.086***		+0.076***	+0.084***
		(+9.81)		(+7.95)	(+8.64)
RiskDensity			−4.743***	−2.526***	
			(−7.08)	(−3.33)	
UncDensity					−0.106
					(−0.11)
LitDensity					−4.709***
					(−5.38)
$\ln \sigma_{i,t}^{\text{pre}}$	+0.541***	+0.505***	+0.525***	+0.501***	+0.498***
	(+22.22)	(+21.05)	(+20.89)	(+20.58)	(+20.48)
ln(Market cap)	−0.037***	−0.039***	−0.038***	−0.040***	−0.037***
	(−6.83)	(−7.14)	(−7.32)	(−7.38)	(−7.06)
Leverage	+0.005	−0.006	−0.006	−0.010	−0.010
	(+0.19)	(−0.20)	(−0.19)	(−0.36)	(−0.34)
ROA	−0.503***	−0.398***	−0.481***	−0.398***	−0.401***
	(−6.10)	(−5.14)	(−6.19)	(−5.27)	(−5.41)
Asset growth	+0.077***	+0.058***	+0.070***	+0.057***	+0.054***
	(+5.70)	(+4.42)	(+5.22)	(+4.27)	(+4.11)
Book-to-market	−0.006	−0.031	−0.015	−0.033	−0.030
	(−0.16)	(−0.90)	(−0.43)	(−0.95)	(−0.87)
Past 12-month return	+0.075**	+0.064*	+0.074**	+0.064*	+0.064*
	(+2.27)	(+1.95)	(+2.19)	(+1.95)	(+1.95)
Current ratio	+0.019***	+0.013***	+0.017***	+0.013***	+0.013***
	(+4.14)	(+3.18)	(+3.68)	(+3.03)	(+2.99)
OCF/assets	+0.075	+0.031	+0.017	+0.005	−0.009
	(+0.63)	(+0.27)	(+0.15)	(+0.04)	(−0.08)
Industry and year FE	Yes	Yes	Yes	Yes	Yes
Observations	5,782	5,782	5,782	5,782	5,782
Firms	512	512	512	512	512
Within R^2	0.575	0.591	0.582	0.593	0.594
Δ within R^2 vs (1)	0.0000	0.0159	0.0066	0.0175	0.0190
Wald $\beta^{\text{unc}} = \beta^{\text{lit}}$, p					0.000

Notes. Dependent variable $\ln \sigma_{i,t+1}^{[365]}$. All columns use the same sample and the common control vector. Industry = two-digit SIC code at the filing date. t-statistics in parentheses use standard errors clustered two-way by firm and fiscal year. Within R^2: share of the variance left after the fixed effects that the regressors explain. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.3 The omission gap (H3)

Table 4.6 reports the H3 specification at six horizons under both fixed-effect structures, for the omission gap and for its length-invariant version.

Table 4.6: H3: omission gap and post-filing volatility

	5 days	10 days	30 days	90 days	180 days	365 days
Omission^{1A}, Industry + year FE						
$\hat{\beta}$	−0.015	−0.032	+0.081	+0.238	+0.333	+0.206
	(−0.03)	(−0.10)	(+0.37)	(+1.21)	(+1.55)	(+1.17)
Holm p	1.000	1.000	1.000	1.000	1.000	–
N	4,261	4,652	4,659	4,659	4,659	4,659
Omission^{1A}, Firm + year FE						
$\hat{\beta}$	+0.418	+0.414	+0.475	+0.639	+0.613	+0.274
	(+0.67)	(+0.49)	(+0.70)	(+1.09)	(+1.17)	(+0.61)
Holm p	1.000	1.000	1.000	1.000	1.000	–
N	4,261	4,652	4,659	4,659	4,659	4,659
Length-invariant omission (3,000-token blocks), Industry + year FE						
$\hat{\beta}$	+0.542	+0.548**	+0.487	+0.452*	+0.443*	+0.285*
	(+1.32)	(+1.97)	(+1.60)	(+1.82)	(+1.90)	(+1.80)
Holm p	1.000	0.491	0.773	0.546	0.516	–
N	4,261	4,652	4,659	4,659	4,659	4,659
Length-invariant omission (3,000-token blocks), Firm + year FE						
$\hat{\beta}$	+0.548	+0.679	+0.351	+0.433	+0.424	+0.097
	(+0.90)	(+1.08)	(+0.58)	(+0.82)	(+1.12)	(+0.34)
Holm p	1.000	1.000	1.000	1.000	1.000	–
N	4,261	4,652	4,659	4,659	4,659	4,659

Notes. Each cell is a separate regression of $\ln \sigma^{[h]}_{i,t+1}$ on the omission measure, $\ln(\text{Words}^{1A})$, RiskDensity and the common controls. Holm p: Holm-adjusted over the ten tests of each measure at $h \in \{5,10,30,90,180\}$ under both FE structures (365 days is outside the family). Two-way clustered t-statistics. Verdict under the decision rule of Section 3.5: not supported. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

At the 30-day horizon the omission coefficient is +0.081 ($t = +0.37$) with industry fixed effects and +0.475 ($t = +0.70$) with firm fixed effects; no coefficient in the family of ten tests is significant even before the Holm adjustment, and all Holm-adjusted p-values equal one. The length-invariant gap gives the same picture (+0.487, $t = +1.60$, and +0.351, $t = +0.58$). Under the decision rule of Section 3.5, **H3 is not supported**: the omission gap is not associated with post-filing volatility once length, density and the controls are held fixed. Chapter 5 shows that no variant of the measure changes this conclusion.

5. Robustness

This chapter checks the results of Chapter 4 against other horizons, specifications, samples and methods of inference, and examines the omission measure of H3 in detail.

5.1 Alternative horizons

Table 5.1 re-estimates the composite and decomposition specifications with volatility over 5, 10, 30, 90, 180 and 365 days. The length coefficient is positive and significant at the 1% level at every horizon (between +0.058 and +0.101 in the decomposition). LitDensity is negative and significant from 30 days on (t between −2.90 and −5.38), but not at 5 or 10 days, where the windows contain few returns. UncDensity is never significant. The composite RiskDensity is significant at the 5% level from 90 days on. Figure 5.1 plots the length and LitDensity coefficients. The coefficients are not compared formally across horizons, because the windows overlap and the estimates are correlated.

Table 5.1: Text coefficients at alternative horizons

	5 days	10 days	30 days	90 days	180 days	365 days
Decomposition specification (column 5)						
ln(Words^{1A})	+0.101***	+0.071***	+0.058***	+0.070***	+0.076***	+0.084***
	(+3.70)	(+4.48)	(+4.54)	(+7.60)	(+7.25)	(+8.64)
UncDensity	−1.353	+0.009	−0.545	+0.442	+0.544	−0.106
	(−0.39)	(+0.00)	(−0.31)	(+0.30)	(+0.49)	(−0.11)
LitDensity	+1.205	−2.017	−3.018***	−4.767***	−4.556***	−4.709***
	(+0.46)	(−1.08)	(−2.90)	(−4.54)	(−5.09)	(−5.38)
Composite specification (column 4)						
ln(Words^{1A})	+0.105***	+0.067***	+0.053***	+0.060***	+0.066***	+0.076***
	(+4.14)	(+4.76)	(+4.51)	(+7.48)	(+6.67)	(+7.95)
RiskDensity	−0.025	−1.056	−1.845*	−2.296**	−2.137**	−2.526***
	(−0.01)	(−0.69)	(−1.70)	(−2.16)	(−2.55)	(−3.33)
Observations	5,300	5,775	5,782	5,782	5,782	5,782
Within R^2 (col. 5)	0.170	0.291	0.368	0.443	0.522	0.594

Notes. Each column re-estimates the specification with $\ln \sigma^{[h]}_{i,t+1}$ as the dependent variable; common controls, industry and year FE. Industry = two-digit SIC code at the filing date. t-statistics in parentheses use standard errors clustered two-way by firm and fiscal year. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

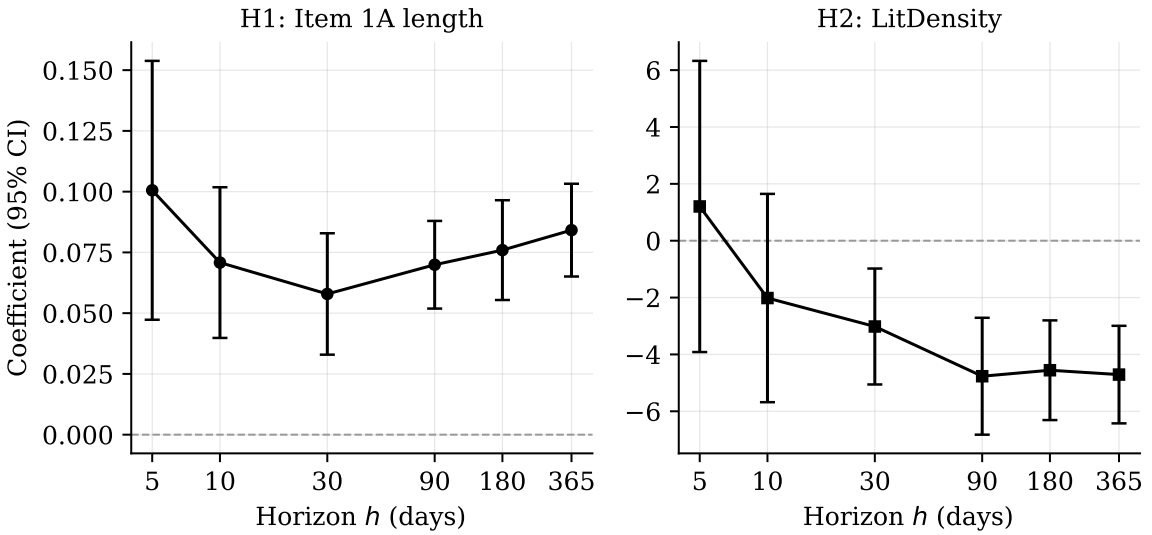


Figure 5.1: Length (left) and LitDensity (right) coefficients with 95% confidence intervals at the six horizons (decomposition specification, industry and year fixed effects; horizons on a log scale).

5.2 Robustness of the H1 and H2 estimates

Table 5.2 reports the decomposition specification under alternative choices, one change per row (Table A.8 in Appendix A repeats it at 365 days).

Market conditions and earnings releases. Adding the market's volatility over the same window, replacing the fiscal-year fixed effects by filing-month × fiscal-year effects, and using idiosyncratic volatility as the dependent variable leave the length coefficient between +0.053 and +0.089 and LitDensity between −2.235 and −3.399, all significant at the 5% level. Earnings releases rarely fall in the 30-day window, because the release usually precedes the 10-K; dropping the windows that contain one, or removing the release day and the following day from every window, does not change the estimates.

Fiscal and calendar time. The filings of one fiscal year span a median of 390 calendar days (Table A.3), so the fiscal-year effects do not absorb calendar-time shocks. Replacing them by effects for the calendar month of the filing (169 months) leaves the length coefficient at +0.057 ($t = +4.94$) and LitDensity at −2.329 ($t = -2.62$). In the sample of the 371 firms with a December fiscal-year end, where fiscal and calendar time coincide (4,200 firm-years), length is +0.060 ($t = +4.64$) and LitDensity −1.906 ($t = -1.96$), significant only at the 10% level: the length result holds in both, while the litigious-density result weakens when the sample is restricted.

Table 5.2: Robustness of the H1/H2 decomposition (30-day window)

	$\ln(\text{Words}^{1A})$	UncDensity	LitDensity	N
Baseline (Table 4.3, col. 5)	+0.058*** (+4.54)	−0.545 (−0.31)	−3.018*** (−2.90)	5,782
+ market volatility in window	+0.067*** (+5.75)	+0.139 (+0.11)	−3.399*** (−4.05)	5,782
Windows with an earnings release dropped	+0.056*** (+4.42)	−0.585 (−0.34)	−3.048*** (−2.96)	5,703
Volatility without earnings-release days	+0.057*** (+4.48)	−0.511 (−0.29)	−3.093*** (−3.03)	5,782
Idiosyncratic volatility as DV	+0.089*** (+6.02)	−2.589 (−1.57)	−3.308*** (−2.61)	5,782
Filing-month × year FE	+0.053*** (+4.61)	+0.020 (+0.01)	−2.235** (−2.37)	5,782
Calendar year-month of filing FE (instead of fiscal year)	+0.057*** (+4.94)	+0.217 (+0.16)	−2.329*** (−2.62)	5,782
Firms with a December fiscal-year end only	+0.060*** (+4.64)	+1.057 (+0.63)	−1.906* (−1.96)	4,200
SIC3 + year FE	+0.065*** (+4.16)	−0.432 (−0.26)	−3.542** (−2.49)	5,782
FF48 + year FE	+0.066*** (+4.89)	−0.944 (−0.56)	−4.052*** (−3.64)	5,782
Today's SIC2 + year FE	+0.058*** (+4.73)	−0.523 (−0.30)	−2.951*** (−2.76)	5,782
Excluding utilities (49) and health services (80)	+0.064*** (+4.91)	−1.406 (−0.70)	−3.005*** (−2.58)	5,112
Excluding weighted-average share counts	+0.047*** (+3.55)	−0.339 (−0.19)	−3.953*** (−3.57)	5,392
Excluding imputed balance-sheet values	+0.078*** (+4.87)	−0.760 (−0.43)	−1.761 (−1.05)	3,820
Balanced panel, FY 2012–2024	+0.081*** (+4.80)	−0.128 (−0.06)	−3.405** (−2.47)	3,367
Firms with ≥ 3 years	+0.057*** (+4.57)	−0.524 (−0.30)	−2.997*** (−2.88)	5,765
Firms with ≥ 5 years	+0.055*** (+4.41)	−0.649 (−0.37)	−3.072*** (−3.01)	5,613
Firms with ≥ 7 years	+0.056*** (+4.31)	−0.702 (−0.39)	−3.181*** (−2.97)	5,432
+ lagged dependent variable	+0.055*** (+4.02)	−0.516 (−0.29)	−2.866*** (−2.77)	5,782
SE clustered by firm only	+0.058*** (+4.87)	−0.545 (−0.40)	−3.018** (−2.19)	5,782
SE clustered by year only	+0.058*** (+4.97)	−0.545 (−0.33)	−3.018*** (−3.90)	5,782
Counts: $\ln(\text{Words})$, $\ln(1+n^{\text{unc}})$, $\ln(1+n^{\text{lit}})$	−0.050* (−1.76)	+0.098** (+2.47)	−0.001 (−0.09)	5,855
Text of year $t-1$ instead of t	+0.057*** (+4.42)	+0.175 (+0.11)	−2.449* (−1.95)	5,345
Level + change: Level $t-1$	+0.063*** (+4.77)	+0.009 (+0.01)	−2.696** (−2.43)	5,289
Level + change: Change $t-1 \to t$	+0.080 (+1.47)	−2.850 (−0.99)	−1.075 (−0.25)	5,289
Post-2019 interaction: FY $\leq$ 2019 (main effect)	+0.046*** (+4.04)	−1.756 (−0.72)	−2.108 (−1.47)	5,782
Post-2019 interaction: × FY $\geq$ 2020	+0.039 (+1.42)	+3.705 (+1.02)	−3.363 (−1.09)	5,782

Notes. Each row is a separate regression of the column-(5) specification with the change described; common controls, industry (historical SIC2) and year FE and two-way clustered standard errors unless stated. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Industry definition. Finer (SIC3) or alternative (Fama–French 48) industry effects, today's rather than the historical SIC code, and excluding utilities (SIC 49) and health services (SIC 80), whose risk disclosures are shaped by regulation, all leave both coefficients significant; with Fama–French industries LitDensity is -4.052 ($t = -3.64$).

Sample. Requiring at least three, five or seven years per firm, using only firms observed in every fiscal year 2012–2024 (3,367 firm-years), and dropping the market capitalisations based on weighted-average share counts leave the results intact. Dropping every firm-year with an imputed balance-sheet value reduces the sample to 3,820 firm-years, mostly because total liabilities are often computed from assets and equity; length stays significant ($+0.078$, $t = +4.87$), but LitDensity is no longer (-1.761, $t = -1.05$). The LitDensity result therefore depends partly on firm-years whose leverage is constructed rather than reported.

Counts instead of ratios. With $\ln(1 + n^{\text{unc}})$ and $\ln(1 + n^{\text{lit}})$ next to $\ln(\text{Words}^{1A})$, no ratio enters the model. The litigious count is negative but significant only at the 10% level ($t = -1.76$), and the uncertainty count is positive ($t = +2.47$). Holding length fixed, a higher count is a higher density, so this specification restates H2 without the ratio; that the litigious coefficient weakens shows that the ratio form matters for its significance.

Timing. Using the text of the previous year's 10-K instead of the current one leaves the length coefficient unchanged ($+0.057$) and LitDensity weaker (-2.449, $t = -1.95$). When the level of year $t - 1$ and the change from $t - 1$ to t enter together, the levels carry the association and the changes are insignificant. The text variables therefore act as slowly moving firm characteristics; the association is not driven by the revisions made in the current filing.

Before and after 2020. Interacting the three text variables with a dummy for fiscal years from 2020 shows no significant change after 2019: the interaction terms have t-statistics of $+1.42$ (length), $+1.02$ (UncDensity) and -1.09 (LitDensity). In this specification the pre-2020 LitDensity coefficient alone is not significant ($t = -1.47$); the pooled estimate is more precise than either sub-period.

Inference with few year clusters. Table 5.3 compares p-values. The wild cluster bootstrap by fiscal year gives $p = 0.001$ for length and $p = 0.005$ for LitDensity in the industry fixed-effects model, close to the two-way clustered values; with one-way firm clustering LitDensity has $p = 0.028$. The conclusions do not depend on the clustering choice.

Table 5.3: Inference with few year clusters (30-day window)

		Coef.	p-value		
FE	Variable		Two-way	Firm	Wild bootstrap (year)
Industry + year	$\ln(\text{Words}^{1A})$ (4)	+0.053	0.000		0.001
Industry + year	RiskDensity (4)	−1.845	0.090		0.092
Industry + year	$\ln(\text{Words}^{1A})$ (5)	+0.058	0.000	0.000	0.001
Industry + year	UncDensity (5)	−0.545	0.757	0.692	0.767
Industry + year	LitDensity (5)	−3.018	0.004	0.028	0.005
Firm + year	$\ln(\text{Words}^{1A})$ (4)	+0.008	0.688		0.687
Firm + year	RiskDensity (4)	−2.727	0.189		0.212
Firm + year	$\ln(\text{Words}^{1A})$ (5)	+0.009	0.654		0.662
Firm + year	UncDensity (5)	−2.372	0.212		0.204
Firm + year	LitDensity (5)	−3.132	0.308		0.349

Notes. Wild cluster bootstrap by fiscal year (15 clusters), 9,999 draws, Webb six-point weights, null imposed, fixed effects partialled out; (4)/(5) = column of Table 4.3. Firm: one-way firm-clustered p (industry FE, column 5).

5.3 Firm fixed effects

The headline models compare firms within an industry and year. Replacing the industry effects by firm effects asks a different question: whether a firm whose Item 1A becomes longer or denser than usual has higher volatility than usual. Table 5.4 reports the answer, and Figure 5.2 compares the coefficients.

Under firm fixed effects the length coefficient falls from +0.058 to +0.009 ($t = +0.45$), and no density coefficient is significant (LitDensity −3.132, $t = -1.02$). Adding the lagged dependent variable, whose coefficient is biased under firm fixed effects (Nickell, 1981), changes nothing (column 3). The associations in Chapter 4 are therefore between-firm patterns: firms with longer and more litigious risk-factor sections differ systematically in subsequent volatility from otherwise similar firms in the same industry and year, but year-to-year changes in a firm's own section do not track changes in its volatility. Two features of the data limit what the firm fixed-effects models can show. First, most of the variation in the text variables is between firms: only 18% of the variance of $\ln(\text{Words}^{1A})$ and 19% of that of LitDensity is within firms (Table 5.5), so the within estimates are imprecise; the confidence interval of the within-firm LitDensity coefficient contains the between-firm estimate. Second, Item 1A changes slowly from year to year (Brown & Tucker, 2011; Cazier et al., 2021), which is consistent with this variance split.

Table 5.4: Industry versus firm fixed effects (30-day window)

	(1) Industry + year	(2) Firm + year	(3) Firm + year, lagged DV
$\ln(\text{Words}^{1A})$	+0.058***	+0.009	+0.015
	(+4.54)	(+0.45)	(+0.65)
UncDensity	−0.545	−2.372	−2.492
	(−0.31)	(−1.25)	(−1.19)
LitDensity	−3.018***	−3.132	−3.327
	(−2.90)	(−1.02)	(−1.05)
$\ln \sigma_{i,t}^{\text{pre}}$	+0.584***	+0.436***	+0.451***
	(+9.85)	(+6.53)	(+8.05)
$\ln \sigma_{i,t}^{[30]}$ (lagged)			−0.099
			(−1.46)
RiskDensity in column-(4) specification	−1.845*	−2.727	−2.883
	(−1.70)	(−1.31)	(−1.31)
Observations	5,782	5,782	5,782
Firms	512	512	512
Within R^2	0.368	0.115	0.125

Notes. All columns include the common control vector and use the same sample. Column (3) adds the lagged dependent variable, which is subject to Nickell bias under firm fixed effects. Standard errors clustered two-way by firm and fiscal year. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5.5: Within-firm and between-firm variation

	N	SD	SD within	Within share	Within share net of year
$\ln \sigma_{i,t+1}^{[30]}$	5,782	0.5794	0.4746	0.671	0.380
$\ln(\text{Words}^{1A})$	5,782	0.6488	0.2748	0.179	0.117
RiskDensity	5,782	0.0065	0.0030	0.213	0.207
UncDensity	5,782	0.0049	0.0022	0.204	0.202
LitDensity	5,782	0.0046	0.0020	0.194	0.179
Omission^{1A}	4,659	0.1001	0.0358	0.128	0.122

Notes. Within share: variance of the deviation from the firm mean over the total variance; the last column first removes fiscal-year means. 30-day estimation sample (omission: H3 sample).

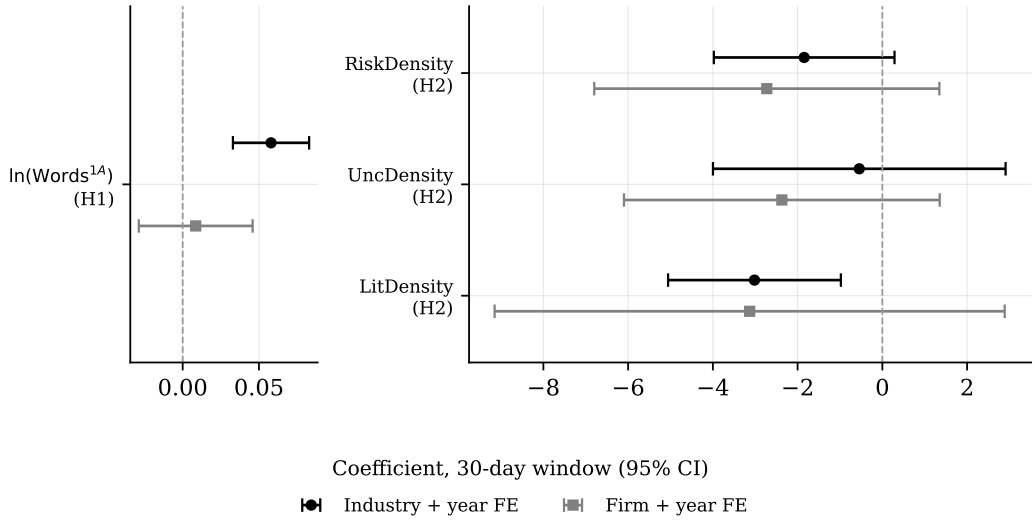


Figure 5.2: Text coefficients at the 30-day horizon with 95% confidence intervals, industry and year fixed effects (circles) versus firm and year fixed effects (squares). The length coefficient falls from +0.058 to +0.009; the density intervals under firm fixed effects are wide and include zero.

5.4 The omission measure (H3)

H3 is not supported in Chapter 4. This section examines the measure itself, to show that the null result does not come from a single construction choice, and reports the checks that address the criticisms of the earlier version of the measure. Table 5.6 reports the coefficients and Table 5.7 the diagnostics.

Length. The omission gap is strongly related to length: $r = -0.915$ with $\ln(\text{Words}^{1A})$ in the H3 sample of 4,659 firm-years (Figure 5.3). A longer section mechanically covers more of the peer vocabulary. Three versions remove this link in different ways. The length-invariant gap computed from blocks of 3,000 tokens has a correlation of +0.424 with length instead (the remaining relation runs through how varied a section's vocabulary is); the residual from a regression of the gap on length within each industry-year cell is uncorrelated with length by construction; and adding $\ln(\text{Words}^{\text{MD\&A}})$ and the square of $\ln(\text{Words}^{1A})$ allows for a non-linear length effect. None of these versions is significant under either fixed-effect structure.

Sources of variation. Under firm fixed effects the gap varies only from year to year. The decomposition in Table 5.7 attributes 91% of the variance of the year-on-year change to changes in the firm's own text, 6% to changes in the peers' texts and 3% to entering and leaving peers. A fixed vocabulary for all years gives a gap with the same (insignificant) coefficients. The within-firm variation of the gap is therefore mostly a revision of the firm's own section, and only 13% of its variance is within firms (Table 5.5).

Table 5.6: H3 robustness (30-day window)

	Firm + year FE		Industry + year FE	
	$\hat{\beta}$ (t)	N	$\hat{\beta}$ (t)	N
Baseline	+0.475 (+0.70)	4,659	+0.081 (+0.37)	4,659
Length-invariant omission	+0.351 (+0.58)	4,659	+0.487 (+1.60)	4,659
Residual on $\ln(\text{Words}^{1A})$ within cell	+0.394 (+0.86)	4,608	+0.025 (+0.08)	4,608
Decile rank within cell (cells ≥ 10)	+0.003 (+0.53)	3,667	+0.002 (+0.57)	3,667
Top-decile dummy (bottom-decile dummy included)	−0.025 (−0.66)	3,667	−0.001 (−0.05)	3,667
Cells with ≥ 20 peers	+0.857 (+1.15)	3,262	+0.233 (+0.93)	3,262
+ $\ln(\text{Words}^{\text{MD\&A}})$ and $\ln(\text{Words}^{1A})^2$	+0.602 (+0.86)	4,659	+0.013 (+0.05)	4,659
Omission of year $t-1$	+0.726** (+1.97)	4,224	+0.223 (+1.37)	4,224
Pooled vocabulary over all years	+0.417 (+1.02)	4,659	+0.084 (+0.38)	4,659
Peers by today's SIC2	+0.702 (+0.92)	4,604	+0.071 (+0.33)	4,604
+ lagged dependent variable	+0.530 (+0.73)	4,659	+0.073 (+0.35)	4,659

Notes. H3 specification with one change per row; coefficient on the (modified) omission measure. Two-way clustered t-statistics. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5.7: H3 diagnostics: placebo, sources of variation, length

Placebo: random SIC2 cells within year, firm + year FE, 30 days	
Permutations	1,000
Observed t	0.70
Placebo t: mean / SD	0.24 / 0.93
Placebo t: 95th / 97.5th percentile	1.69 / 1.96
Empirical p (one-sided / two-sided)	0.315 / 0.466
Decomposition of ΔOmission (n = 4,802)	
Own-text change: share of variance	0.907
Peer-text change: share of variance	0.063
Peer-composition change: share of variance	0.026
Interaction: share of variance	0.078
Correlation with $\ln(\text{Words}^{1A})$, H3 sample	
Omission^{1A}	-0.915
Length-invariant omission	0.424
Pooled-vocabulary omission	-0.933
N	4,659

Notes. Decomposition in the year-t vocabulary using fiscal-year documents: own $= \Omega(o_t, C_{t-1}) - \Omega(o_{t-1}, C_{t-1})$; peer text = change for the peers present in both years; composition = effect of entering/leaving peers; interaction = remainder.

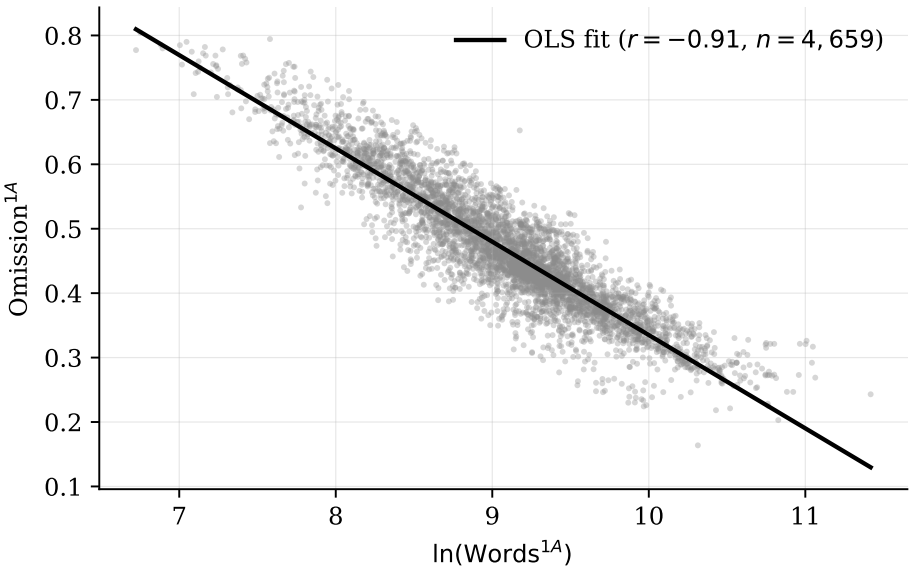


Figure 5.3: Omission gap against ln(Words^{1A}), one point per firm-year of the 30-day H3 sample, with the OLS fit.

Peer definition. Peers defined by today's SIC code, and only cells with at least 20 peers (3,262 firm-years; Table A.9 in Appendix A lists the industries, which are dominated by business services, utilities, chemicals and pharmaceuticals, machinery and electronics), give the same result.

Ranks and tails. The decile rank of the gap within its industry-year cell and a dummy for the top decile (with the bottom decile controlled) are insignificant. These versions need cells with at least ten firms, which reduces the sample from 4,659 to 3,667 firm-years; 992 firm-years lie in smaller cells.

Timing. The gap of the previous fiscal year, which is known long before the current filing and cannot reflect conditions at that filing, has a positive coefficient that is significant at the 5% level under firm fixed effects (+0.726, $t = +1.97$) but not under industry effects ($t = +1.37$). As one of many checks, and without a significant contemporaneous counterpart, this does not change the conclusion.

Placebo. Firms were reassigned at random to industry cells within each fiscal year, keeping the cell sizes, the gap was recomputed and the firm fixed-effects model re-estimated, 1,000 times. The observed t-statistic of +0.70 lies in the middle of the placebo distribution: the empirical two-sided p-value is 0.466, and 2.4% of the placebo t-statistics exceed 1.96 (Figure 5.4). The real peer groups carry no more information about volatility than random ones.

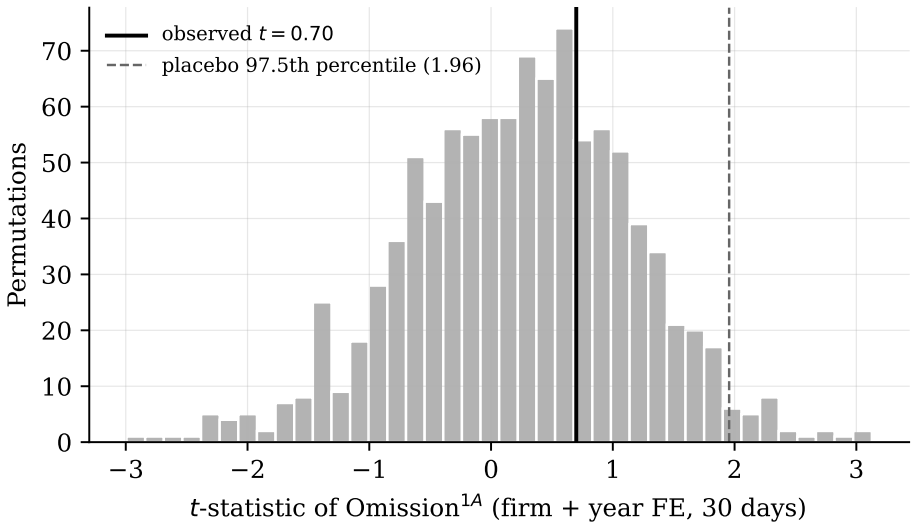


Figure 5.4: Distribution of the omission-gap t-statistic in 1,000 placebo assignments of firms to industry cells (firm and year fixed effects, 30-day window), with the observed value.

Summary for H3. Across the eleven specifications and both fixed-effect structures, the omission coefficient is significant in one (the prior-year gap under firm fixed effects, at the 5% level without a multiple-testing adjustment). The verdict of Section 4.3 stands: H3 is not supported.

5.5 Persistence and unit roots

With at most 15 annual observations per firm, the main risk of a dynamic panel is the Nickell bias discussed in Section 3.6, not a spurious regression. For completeness, Table 5.8 reports two panel unit-root tests designed for short panels. Every variable is first demeaned by fiscal year. The Harris–Tzavalis test (Harris & Tzavalis, 1999) uses the balanced panel of firms observed in all years 2012–2024 and tests the null that every firm's series has a unit root against a common autoregressive coefficient below one; its p-values are simulated for the actual number of firms and years. The Im–Pesaran–Shin test (Im et al., 2003) averages firm-level augmented Dickey–Fuller statistics on the unbalanced panel; its null is that all firms have a unit root, and a rejection means that at least some firms' series are stationary, not that all are.

Both tests reject the unit-root null for volatility, the text variables and most controls. Two variables are exceptions. For leverage the Harris–Tzavalis test does not reject ($\hat{\rho} = 0.772$, $p = 0.592$), while the Im–Pesaran–Shin test does; for the omission gap neither test rejects at the 5% level ($\hat{\rho} = 0.758$, $p = 0.257$; IPS $p = 0.099$). Leverage is a control, and the omission gap enters only the H3 models, whose coefficient is insignificant. A ratio bounded between zero and one cannot follow a random walk forever, but over 13 years it can be persistent enough that these tests lack power; the results are therefore read as showing high persistence

Table 5.8: Panel unit-root tests

	Harris–Tzavalis (balanced, 2012–2024)			Im–Pesaran–Shin (1 lag)		
	N	$\hat{\rho}$	p	N	$W_{\bar{t}}$	p
$\ln \sigma_{i,t+1}^{[30]}$	379	-0.077	0.000	449	-16.47	0.000
$\ln \sigma_{i,t}^{\text{pre}}$	379	0.180	0.000	449	-16.29	0.000
ln(Market cap)	371	0.460	0.000	437	-3.76	0.000
Leverage	344	0.772	0.592	437	-3.67	0.000
ROA	358	0.419	0.000	433	-9.11	0.000
Asset growth	378	-0.052	0.000	449	-20.82	0.000
Book-to-market	362	0.516	0.000	430	-6.89	0.000
Past 12-month return	379	-0.108	0.000	449	-26.48	0.000
Current ratio	368	0.446	0.000	439	-10.10	0.000
OCF/assets	377	0.437	0.000	448	-9.49	0.000
ln(Words^{1A})	339	0.351	0.000	417	-14.07	0.000
RiskDensity	321	0.673	0.000	402	-4.40	0.000
UncDensity	321	0.659	0.000	402	-5.36	0.000
LitDensity	321	0.697	0.000	402	-1.92	0.027
Omission^{1A}	220	0.758	0.257	335	-1.29	0.099

Notes. Variables cross-sectionally demeaned by fiscal year. Harris–Tzavalis: within estimate of ρ on firms observed in every year 2012–2024 ($T = 13$); H_0: $\rho = 1$ for all firms; p from 2,000 simulated random-walk panels of the same N and T. IPS: mean of firm ADF t-statistics (intercept, one lag; firms with ≥ 8 consecutive years), standardised with simulated null moments; H_0: all firms have a unit root, H_1: some firms are stationary.

for these two variables rather than as proof of stationarity.

Conclusion

This thesis asked whether three properties of the risk-factor section of the 10-K, its length, its share of uncertainty and litigious words, and what it leaves out relative to industry peers, are associated with stock-return volatility in the weeks after the filing. The tests use a panel of 5,782 firm-years of 512 large U.S. non-financial firms for fiscal years 2010–2024, built from SEC filings, and control for the volatility before the filing, firm size, valuation, past returns and accounting fundamentals.

Findings. *H1 is supported between firms.* Firms with longer Item 1A sections have higher 30-day post-filing volatility than otherwise similar firms in the same industry and year: one standard deviation of log length is associated with 3.8% higher volatility. The association holds at every horizon from 5 to 365 days, under alternative industry definitions, with market and idiosyncratic volatility, and with a wild cluster bootstrap. It does not hold within firms: when a firm's own section becomes longer, its volatility does not rise with it.

H2 is not supported at the 30-day horizon. Holding length fixed, the composite density of uncertainty and litigious words has a negative but insignificant coefficient at 30 days; it is negative and significant only at horizons of 90 days and longer. Of its two parts, the uncertainty density has no association with volatility at any horizon, while the litigious density is negative and significant from 30 days on, but this result weakens when firm-years with imputed balance-sheet values are excluded and disappears within firms. The negative sign is more in line with litigious vocabulary acting as standard cautionary language than with it signalling additional risk, but the evidence for H2 is weak.

H3 is not supported. The share of the peers' vocabulary that a firm does not use in its Item 1A is not associated with subsequent volatility under industry or firm fixed effects, after a Holm adjustment over horizons and fixed-effect structures, in a length-invariant version of the measure, or relative to 1,000 random peer assignments. The measure is dominated by section length, and its year-to-year variation comes mainly from the firm's own revisions. The data give no support for a strategic-silence interpretation of omissions.

Together, the results suggest that the volume of risk disclosure is the most robust textual correlate of post-filing volatility among the measures studied, and that it reflects persistent differences between firms rather than year-to-year changes in what a firm discloses. Once pre-filing volatility is controlled for, the text variables add little explanatory power: the within-industry-year R^2 rises from 0.363 to 0.368.

Limitations. The results are associations, not causal effects. The firm list was taken from the current SEC ranking, so firms that failed or were acquired during the sample period are missing and early years contain only firms that later became large. Industry peers are defined

by two-digit SIC codes, which group firms with different businesses; alternative industry definitions do not change the H1 and H2 results, but the omission measure depends on the peer definition. With at most 15 years per firm and little within-firm variation in the text variables, the firm fixed-effects estimates are imprecise. The text extraction was audited by two language models on a stratified sample of 200 sections and checked by hand on 51 of them; about 5–7% of sections are deficient, mostly at their boundaries and in early years, and the share of sections taken from exhibits rather than the main document was not recorded. The dictionary densities measure the vocabulary of a section, not its specificity.

Further research. The finding that length matters between but not within firms invites a closer look at what makes some firms' disclosures persistently longer, for example their business complexity or litigation history. Measures of disclosure content that do not depend on fixed word lists, and peer groups based on product descriptions rather than SIC codes (Hoberg & Phillips, 2016), could test the omission idea with less noise. A broader, point-in-time sample including firms that later left the market would show whether the results extend beyond large surviving firms.

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Appendices

A. Data Construction and Supplementary Tables

This appendix collects the tables on sample construction and data quality referred to in Chapters 2–5. Definitions, sources and timing of all variables are documented in the file `docs/data_dictionary.md` of the accompanying repository.

A.1 Sample construction and coverage

Table A.1: Sample construction

Stage	Firms	Firm-years
Universe (SEC ticker list, reconstructed with current metadata)		
SEC ticker list, first 1,000 rows	1,000	
− not an operating company	753	
− financial (SIC 6000-6999)	600	
− no fiscal-year end	599	
− < 5 years of 10-K XBRL data	550	
Frozen universe (548 firms) to panel		
≥ 5 years of 10-K XBRL data	548	8,000
transitions & dedup	548	7,958
drop earliest year	548	7,410
cap FY ≤ 2024	548	6,757
Panel to 30-day estimation sample		
valid 30d post-filing volatility	538	6,603
pre-filing liquidity rule	538	6,598
text from the panel's 10-K	538	6,592
Item 1A text (≥ 100 tokens)	528	6,387
non-missing pre_log_vol	528	6,387
non-missing log_mcap	522	6,253
non-missing leverage	521	6,083
non-missing roa	520	5,934
non-missing asset_growth	520	5,919
non-missing bm	520	5,919
non-missing ret_pre_12m	520	5,919
non-missing current_ratio	512	5,787
non-missing ocf_to_assets	512	5,782

Notes. The 1,000 firms are the first 1,000 rows of the SEC `company_tickers.json` list as stored in the repository (2026-10-12); the universe file holds the 548 firms that passed these filters in the original run. Each estimation-sample row applies one more requirement.

Table A.2: Text coverage by fiscal year

Fiscal year	Filings	Item 1A (%)	Item 7 (%)
2010	152	92.8	90.1
2011	313	94.2	92.0
2012	403	95.3	92.6
2013	424	95.5	93.4
2014	430	96.7	94.4
2015	442	95.9	94.1
2016	460	96.3	94.6
2017	473	96.0	94.5
2018	484	97.5	96.1
2019	496	97.2	95.8
2020	514	96.7	95.9
2021	532	97.4	96.6
2022	542	97.2	96.5
2023	547	97.6	96.9
2024	545	94.7	93.9
All	6,757	96.3	94.9

Notes. Share of panel firm-years whose stored section has at least 100 tokens.

Table A.3: Fiscal-year ends and filing dates in the estimation sample (30-day window)

Month	Fiscal year ends (%)	10-K filed (%)
Jan	5.0	2.8
Feb	0.0	66.5
Mar	2.4	8.7
Apr	1.2	0.6
May	1.7	2.3
Jun	5.4	1.6
Jul	1.6	1.5
Aug	1.2	5.2
Sep	5.2	1.6
Oct	2.7	1.4
Nov	1.0	5.0
Dec	72.6	2.7

Notes. Shares of the 5,782 firm-years. Fiscal-year end month: month of the period end, periods ending on 1–7 January counted to December. The median lag between period end and filing is 53 days (10th percentile 38, 90th percentile 59). 145 of the 512 firms have a fiscal year that does not end in December. 73.6% of the 10-Ks are filed in the calendar year after the fiscal year, and the filings of one fiscal year span a median of 390 calendar days; the filings fall in 169 distinct calendar months.

A.2 XBRL tags

Table A.4: XBRL tags: choices, locking losses and overlap

Metric	Locked tag	
Net income	`ProfitLoss`	
	`NetIncomeLoss`	
	`IncomeLossFromContinuingOperations`	
Operating cash flow	`NetCashProvidedByUsedInOperatingActivities`	
	`NetCashProvidedByUsedInOperatingActivitiesContinuingOperations`	
Stockholders' equity	`StockholdersEquityIncludingPortionAttributableToNoncontrollingInterest`	
	`StockholdersEquity`	
Values set to missing by per-firm tag locking		
Stockholders' equity	637 of 7,900	171 firms with >
Net income	38 of 7,813	209 firms with >
Operating cash flow	0 of 7,956	217 firms with >
Paired tags reported together: |difference| / total assets (pp)		
Net income	n = 3,838; median 0.043; p95 1.48; p99 3.56; max 37.4	>1%
Operating cash flow	n = 417; median 0.035; p95 1.80; p99 4.19; max 6.5	>1

Notes. Net-income pair: `ProfitLoss` (consolidated) vs `NetIncomeLoss` (parent); their difference is the non-controlling interest, and the panel uses the consolidated figure. Operating-cash-flow pair: total vs continuing operations. ">1%": differences above 1% of total assets. Locking: for metrics with non-equivalent tags, each firm keeps its most frequent tag.

A.3 Filing-frequency pilot tests

The following tables come from the pilot tests described in Section 2.2. They cover 30 firms (financial data) and 15 firms (text) and are illustrations, not estimates for the panel.

Table A.5: XBRL Financial Metric Coverage: Annual (10-K) vs. Quarterly (10-Q)

Metric	Annual (%)	Quarterly (%)
assets_current	87.0	91.1
eps_diluted	89.4	92.7
liabilities_current	87.0	91.1
net_income	93.6	97.3
operating_cash_flow	89.2	35.8
operating_income	76.9	80.5
stockholders_equity	98.7	94.9
total_assets	89.9	93.3
total_liabilities	63.7	66.5
Median periods per firm	20	52
Firms with data	30	30

Note: Coverage shows the percentage of firm-periods with non-missing values for each metric. Annual data is extracted from 10-K filings; quarterly data from 10-Q filings (Q1–Q3 only). Sample: 30 non-financial operating companies from the top 200 by market capitalisation.

Table A.6: Annual vs. Quarterly Approach: Summary Comparison

Criterion	Annual (10-K)	Quarterly (10-Q)
Filing type	10-K	10-Q (Q1–Q3 only)
MD&A availability	93% (Item 7)	100% (Part I, Item 2)
Risk Factors availability	100% (Item 1A)	∼52% substantive
Risk Factors content	Median ∼30,000 words	Median ∼9,000 words (substantive); 48% boilerplate
Q4 financial data	Included in 10-K	Unreliable / absent
Explicit Q4 XBRL facts	N/A	3–43% of firm-years
Implied Q4 mismatch rate	N/A	∼11% of comparisons
Cash-flow quarterly coverage	N/A	7% of firm-years
Item 1A length (H1)	Feasible (annual word count)	Limited (52% substantive)
Item 1A risk-word density (H2)	Feasible (annual LM ratio)	Limited (52% substantive)
Literature alignment	Standard	Uncommon
Expected observations	∼6,400 firm-years	∼19,000 firm-quarters

Note: Summary based on diagnostic tests of 30 non-financial companies (XBRL, Q4 tests) and 15 companies (textual tests). Risk Factors in 10-Q filings are required in Form 10-Q (Part II, Item 1A) only for "material changes" from the most recent 10-K; roughly half of companies include substantive risk disclosures, while the rest provide only a boilerplate reference. See Tables A.5, 2.1, and 2.2 for detailed results.

Table A.7: Per-Company 10-Q Textual Content (Most Recent 6 Filings)

Ticker	Filings	MD&A Found	MD&A Med. Words	Risk Subst.	Risk Boilerplate
AAPL	6	6	2,781	3	3
AMZN	6	6	6,291	6	0
AVGO	6	6	4,778	6	0
COST	6	6	4,979	0	6
GOOGL	6	6	8,096	3	3
JNJ	6	6	7,587	0	6
LLY	6	6	4,994	0	6
MA	6	6	7,300	6	0
META	6	6	7,756	6	0
MSFT	6	6	7,981	0	6
NVDA	6	6	5,156	6	0
TSLA	6	6	5,582	2	4
V	6	6	5,043	0	6
WMT	6	6	7,142	3	3
XOM	6	6	6,197	6	0

Note: "MD&A Found" and "Risk Subst." count filings with >200 substantive words. "Risk Boilerplate" counts filings where the risk-factors section contains only a reference to the annual 10-K.

A.4 Additional robustness tables

Table A.8: Robustness of the H1/H2 decomposition (365-day window)

	$\ln(\text{Words}^{1A})$	UncDensity	LitDensity	I
Baseline (Table 4.3, col. 5)	+0.084*** (+8.64)	−0.106 (−0.11)	−4.709*** (−5.38)	5,78
+ market volatility in window	+0.085*** (+8.47)	−0.189 (−0.21)	−4.641*** (−5.32)	5,78
Volatility without earnings-release days	+0.069*** (+7.07)	−0.426 (−0.44)	−4.087*** (−4.53)	5,78
Idiosyncratic volatility as DV	+0.107*** (+8.88)	−2.468** (−2.17)	−5.227*** (−4.72)	5,78
Filing-month × year FE	+0.077*** (+8.36)	+0.043 (+0.05)	−4.046*** (−5.42)	5,78
Calendar year-month of filing FE (instead of fiscal year)	+0.078*** (+8.52)	+0.064 (+0.07)	−3.958*** (−5.20)	5,78
Firms with a December fiscal-year end only	+0.079*** (+8.12)	+0.677 (+0.70)	−2.608** (−2.47)	4,20
SIC3 + year FE	+0.080*** (+7.79)	−0.299 (−0.28)	−4.138*** (−3.93)	5,78
FF48 + year FE	+0.083*** (+9.02)	−0.020 (−0.02)	−4.958*** (−5.70)	5,78
Today's SIC2 + year FE	+0.086*** (+9.48)	−0.166 (−0.18)	−4.500*** (−5.28)	5,78
Excluding utilities (49) and health services (80)	+0.085*** (+7.59)	−0.713 (−0.73)	−4.421*** (−4.72)	5,11
Excluding weighted-average share counts	+0.080*** (+7.30)	−0.366 (−0.37)	−4.858*** (−4.91)	5,39
Excluding imputed balance-sheet values	+0.087*** (+6.27)	−1.494 (−1.30)	−4.332*** (−3.37)	3,82
Balanced panel, FY 2012–2024	+0.097*** (+8.25)	+1.105 (+0.89)	−4.807*** (−4.08)	3,36
Firms with ≥ 3 years	+0.084*** (+8.58)	−0.134 (−0.14)	−4.731*** (−5.42)	5,76
Firms with ≥ 5 years	+0.084*** (+8.27)	−0.168 (−0.18)	−4.699*** (−5.26)	5,61
Firms with ≥ 7 years	+0.082*** (+8.05)	−0.203 (−0.20)	−4.598*** (−4.98)	5,43
+ lagged dependent variable	+0.062*** (+7.42)	−0.181 (−0.27)	−3.495*** (−4.94)	5,78
SE clustered by firm only	+0.084*** (+8.31)	−0.106 (−0.10)	−4.709*** (−4.71)	5,78
SE clustered by year only	+0.084*** (+14.10)	−0.106 (−0.17)	−4.709*** (−9.12)	5,78
Counts: $\ln(\text{Words})$, $\ln(1+n^{\text{unc}})$, $\ln(1+n^{\text{lit}})$	−0.079*** (−4.80)	+0.164*** (+7.37)	−0.015 (−1.25)	5,85
Text of year $t-1$ instead of t	+0.080*** (+7.74)	+0.117 (+0.13)	−4.103*** (−4.92)	5,34
Level + change: Level $t-1$	+0.085*** (+8.24)	+0.107 (+0.11)	−4.340*** (−4.79)	5,28
Level + change: Change $t-1 \to t$	+0.067*** (+2.69)	−0.949 (−0.46)	−2.456 (−1.05)	5,28
Post-2019 interaction: FY $\leq$ 2019 (main effect)	+0.067*** (+5.19)	−1.597 (−1.51)	−4.886*** (−4.76)	5,78
Post-2019 interaction: × FY $\geq$ 2020	+0.052** (+2.43)	+4.418** (+2.40)	−0.164 (−0.12)	5,78

Notes. Each row is a separate regression of the column-(5) specification with the change described; common controls, industry (historical SIC2) and year FE and two-way clustered standard errors unless stated. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.9: Industries in the $\geq$ 20-peer H3 sample

SIC2	Firm-years	Firms	Share	Share of full H3 sample
73	754	83	0.231	0.164
49	527	46	0.162	0.113
28	486	48	0.149	0.108
35	455	44	0.139	0.102
38	431	40	0.132	0.095
36	407	39	0.125	0.091
37	202	21	0.062	0.053

Notes. SIC2 = two-digit SIC code at filing (e.g. 28 chemicals and pharmaceuticals, 35 industrial machinery and computers, 36 electronic equipment, 38 instruments, 49 utilities, 73 business services).

B. Documentation of AI Tool Usage

This appendix documents the use of artificial intelligence tools in preparing this thesis, as required by the university.

Extraction audit 1 (Section 2.4.4)

Tool. Google Gemini 2.5 Flash, accessed through the OpenAI-compatible Gemini API, temperature 0.

Input and task. The model received one stored section at a time (the full extracted text of Item 1A or Item 7 with its word count, ticker and fiscal year) and a five-criterion rubric (correct section, start boundary, end boundary, content quality, completeness) with three levels (Pass, Minor, Fail). It did not receive the rest of the filing, so boundary judgements are made from the extracted text alone. The prompt is part of `scripts/validate_extraction.py`.

Sample. 200 sections, drawn at random within five three-year fiscal-year strata (2010–2012, . . . , 2022–2024), 20 per stratum and section type; Item 1A and Item 7 were drawn independently (197 distinct filings). The texts the model rated are identical to the stored texts used in the analysis.

Extraction audit 2

Tool. Claude Opus 5.5 (Anthropic), used through the coding assistant described below, on 1 November 2026.

Input and task. The same 200 sections and the same rubric, blind to the ratings of audit 1. Because of the length of the sections, the model received for each section the first and last 1,200 characters, a 600-character excerpt from the middle and the word count. It therefore judges the start and end boundaries on the same evidence as audit 1 but sees less of the body of each section. The ratings are stored in the repository (`admin/claude_audit/`).

Results. Table B.1 reports both audits. The two models agree on the overall verdict for 96.5% of the sections.

Table B.1: Extraction audits: sections rated acceptable ($n = 200$ sections, 100 Item 1A and 100 Item 7)

	Gemini 2.5 Flash (full section)			Claude Opus 5.5 (excerpts)		
Criterion	Strict %	Lenient %	95 % CI	Lenient %	Agreement	κ
Correct section	96.0	96.0	[92.3, 98.0]	97.0	99.0	0.85
Start boundary	95.5	96.5	[93.0, 98.3]	96.5	100.0	1.00
End boundary	96.0	96.5	[93.0, 98.3]	96.0	98.5	0.79
Content quality	95.5	97.5	[94.3, 98.9]	97.5	100.0	1.00
Completeness	94.5	95.5	[91.7, 97.6]	97.5	98.0	0.70
Overall (all five)	89.5	93.5	[89.2, 96.2]	95.0	96.5	0.68
Gemini: sections with at least one Fail, by fiscal-year stratum						
FY 2010–2012	5 of 40					
FY 2013–2015	3 of 40					
FY 2016–2018	2 of 40					
FY 2019–2021	2 of 40					
FY 2022–2024	1 of 40					
Manual check by the author (51 sections, correct section and both boundaries)						
Correct section	0 deficient; agrees with Gemini 94.1%			agrees with Claude 98.0%		
Start boundary	2 deficient; agrees with Gemini 100.0%			agrees with Claude 100.0%		
End boundary	2 deficient; agrees with Gemini 100.0%			agrees with Claude 98.0%		
All three	4 deficient; agrees with Gemini 100.0%			agrees with Claude 98.0%		

Notes. Strict = rated Pass; lenient = Pass or Minor issue. The sample is stratified into five three-year fiscal-year strata with 20 sections per stratum and section type; Item 1A and Item 7 were drawn independently (197 distinct filings). 5 sections with no extracted text are counted as Fail on every criterion. Gemini saw the whole extracted section; Claude saw its first and last 1,200 and a middle 600 characters and the word count, blind to the Gemini verdicts. Neither saw the surrounding filing. Agreement and Cohen's κ on acceptable (1) versus Fail (0). Wilson score intervals. Both audits fail the same 8 sections on the overall criterion; 5 fail only under Gemini and 2 only under Claude. Manual check: error rate on the 50 random sections 6.0% (Wilson 2.1–16.2%), against the original filings, blind to both audits.

Limitations. Both audits are LLM judgements of the extracted text without the original filing, and the second model is used as a check of the first, not as ground truth. The author checked 51 sections by hand against the filings on EDGAR, blind to both audits (50 random sections and the remaining LLM-flagged section; labels in `admin/human_labels_template.csv`). The manual verdict on correct section, start and end boundary agrees with audit 1 on all 51 sections and with audit 2 on 50; the four sections the author marked as deficient are the four that audit 1 failed on these criteria. The audits do not compare the parser with other extraction methods, so no claim is made about its accuracy relative to plain-text or XBRL-based extraction.

Code development, analysis and text revision

Tools. AI code completion in VS Code during the original development; for the revision of the thesis, an AI coding agent in VS Code (GitHub Copilot with Claude Opus 5.5).

Purpose. The agent was used to rewrite and test the data and estimation scripts, to re-run the analysis, to generate the tables and figures from the stored results, to check the references against the Crossref database, and to draft revised text of Chapters 2 to 5, the appendices and the conclusion.

Contribution and verification. The research questions, the hypotheses, the decisions on the revision and the decision rule for H3 are the author's. All code, results and text produced with the agent were reviewed by the author; every number in the text is generated by the scripts in the repository, and a script checks the numbers in the text against the generated results.

Writing assistance

Tool. AI writing assistant integrated in VS Code.

Purpose. Grammar and style suggestions for English prose.

Verification. Every suggestion was reviewed before acceptance.